The Lambert W Function

A Real-Variable Guide to Its Branches, Identities, Series, Asymptotics, Bounds, Approximations, and Computation

Consolidated edition merging four independent treatments

Consolidated 28 August 2026

Abstract

The Lambert W function is the inverse of the elementary-looking map $w \mapsto we^w$. It is the natural language for equations in which an unknown occurs both outside and inside an exponential. This article develops the function from first principles, at a level intended to be accessible after a standard sophomore course in calculus and a first course in proof. Special attention is paid to the two real branches: the principal branch W_0 and the nonprincipal branch W_{-1}. We prove their domains, ranges, monotonicity, curvature, derivative formulas, integral identities, power series, square-root expansions at the common branch point, complete logarithmic asymptotic expansions, rigorous elementary bounds, and branch-safe Newton iterations. We also derive rational and continued fraction approximations, contour and real integral representations, and closed forms for representative transcendental equations.

The supplied source archive contains the foundational paper of Corless, Gonnet, Hare, Jeffrey, and Knuth [5], the NIST Digital Library of Mathematical Functions entry [7], papers on analytic approximations, continued fractions, certified high-precision evaluation, bounds for W_{-1}, Schroeder-type series, and applications in circuits, biochemical kinetics, ecology, and prime asymptotics. Those sources inform the presentation and bibliography. All principal results used in the exposition are proved here. Results that are mentioned only as historical or bibliographical context are explicitly labelled as such. A fully commented Python program supplied with the article reproduces the numerical figures.

This consolidated edition merges three further independently written treatments of the same material: their shared core agreed with the text theorem for theorem and constant for constant, and their unique layers — the complete power-tower convergence theorem, inverse-Taylor corrections, the branch-exchange involution, a transcendence theorem, a practitioner's toolkit, additional worked applications, and the r-Lambert outlook — are preserved in a dedicated complements section, with a four-way concordance appendix and a note on the role of W_{-1} in the surrounding Fabius–Rvachev corpus.

Provenance

This volume is the 2026-08-28 editorial consolidation of four independently written treatments of the Lambert W function that arrived together in the corpus inbox. The body of the volume is the most complete of the four; the unique layers of the other three are merged in Section 21, and the four-way agreement record is Appendix G. The verification material of all four — figures, data, and deterministic scripts — is preserved under `assets/`; the absorbed source documents themselves are deleted once merged (git history is the archive, pinned by the SHA-256 hashes below). The pre-crosswalk publication checkpoint was a 66-page A4 artifact of 1,107,064 bytes. The present edition incorporates the later crosswalk edits.

- `Lambert_W_Function_Article/Lambert_W_Function_Article.tex` (the spine)
  SHA-256 `5bca61c4e906ca8a92f283bc92913355ccd4aee26d65112b41474fd16862c6ba`

- `Lambert-W-Article/lambert_w_monograph.tex` (the monograph treatment)
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- `Lambert_W_article/lambert_w_article.tex` (the third treatment)
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  SHA-256 `4b732dbafdb9c4c48ed999a1cfe4b9b06fb39fd248bb8fa6aa4cf4d0365c8302`

Contents

1 Why a new function is needed

Elementary algebra teaches us how to undo addition, multiplication, integer powers, and exponentials. Difficulty appears when the unknown occupies two different roles at once. For example,

$$y\mathrm{e}^{y} = 7, \qquad \mathrm{e}^{3t} = 5t + 2, \qquad s + \log s = 10$$

cannot generally be solved by a finite combination of the usual elementary functions. The first equation is so basic that it is sensible to name its inverse.

Definition 1.1 (Lambert W function)**.** *A value w of the Lambert W function at z is any solution of*

$$w\mathrm{e}^{w} = z. \tag{1}$$

Thus

$$w = W(z) \iff w\mathrm{e}^{w} = z.$$

Because the complex exponential is periodic and the map $w \mapsto w\mathrm{e}^{w}$ is not one-to-one, W has branches. The branches are denoted W_k, where $k \in \mathbb{Z}$. On the real line only W_0 and W_{-1} can take real values.

The symbol W is not an unexplained trick: it records an inverse operation, just as $\sqrt{}$, $\log$, and $\arcsin$ do. Once the branch has been selected, every manipulation must be checked against (1). This simple habit prevents most branch errors.

1.1 A first numerical example

The equation $w\mathrm{e}^{w} = 1$ has a unique real solution

$$\Omega := W_0(1) = 0.567143290409783872999968\ldots,$$

sometimes called the *omega constant*. Indeed, $\Omega\mathrm{e}^{\Omega} = 1$, or equivalently $\Omega = \mathrm{e}^{-\Omega}$. The number is not mysterious once one sees it as the fixed point of $w \mapsto \mathrm{e}^{-w}$.

1.2 Reading guide and prerequisites

Sections 2–5 require only calculus. The power-series sections use coefficient comparison and a proved version of Lagrange inversion. The asymptotic section introduces unsigned Stirling numbers of the first kind and proves the coefficient formula. The integral representation section uses a short, self-contained application of residues; it can be skipped without affecting the real-variable narrative. Numerical methods are developed from the mean value theorem and elementary logarithm inequalities.

Throughout, $\log$ means the real natural logarithm when its argument is positive. The complex principal logarithm is written Log. At the point $-1/\mathrm{e}$, both real branches equal -1. The notation $W_{-1}(0) = -\infty$ is occasionally useful as an *extended* endpoint convention, but $W_{-1}(0)$ is not a finite real number.

2 The two real branches

Everything begins with the graph of

$$f(w) = w\mathrm{e}^{w}.$$

Its derivative and second derivative are

$$f'(w) = \mathrm{e}^{w}(1 + w), \qquad f''(w) = \mathrm{e}^{w}(2 + w).$$

The derivative vanishes only at $w = -1$, where

$$f(-1) = -\frac{1}{\mathrm{e}}.$$

Theorem 2.1 (Real branches, domains, and ranges). *The real solutions of $w\mathrm{e}^w = x$ are as follows.*

(i) *If $x < -1/\mathrm{e}$, there is no real solution.*

(ii) *If $x = -1/\mathrm{e}$, there is one real solution, $w = -1$.*

(iii) *If $-1/\mathrm{e} < x < 0$, there are exactly two real solutions:*

$$W_0(x) \in (-1, 0), \qquad W_{-1}(x) \in (-\infty, -1).$$

(iv) *If $x \geq 0$, there is exactly one real solution, $W_0(x) \in [0, \infty)$.*

More precisely,

$$W_0 : [-1/\mathrm{e}, \infty) \longrightarrow [-1, \infty)$$

is a strictly increasing bijection, while

$$W_{-1} : [-1/\mathrm{e}, 0) \longrightarrow (-\infty, -1]$$

is a strictly decreasing bijection.

Proof. Because $\mathrm{e}^w > 0$, the sign of $f(w) = w\mathrm{e}^w$ is the sign of w. Moreover,

$$f'(w) < 0 \quad (w < -1), \qquad f'(w) > 0 \quad (w > -1).$$

Thus f is strictly decreasing on $(-\infty, -1]$ and strictly increasing on $[-1, \infty)$. The endpoint limits are

$$\lim_{w\to-\infty} w\mathrm{e}^w = 0^-, \quad f(-1) = -1/\mathrm{e}, \quad f(0) = 0, \quad \lim_{w\to\infty} w\mathrm{e}^w = \infty.$$

Consequently the restriction of f to $[-1, \infty)$ maps that interval bijectively onto $[-1/\mathrm{e}, \infty)$, and its inverse is W_0. The restriction to $(-\infty, -1]$ maps that interval bijectively, with reversed orientation, onto $[-1/\mathrm{e}, 0)$, and its inverse is W_{-1}. The listed numbers of solutions follow immediately. □

Remark 2.2 (Formalized real-branch package). The project records the exact nonpositive principal-branch image

$$W_0\big([-1/\mathrm{e}, 0]\big) = [-1, 0]$$

as `principalLambertW_image_Icc` in `PrincipalLambertW.lean`. Together with the lower-branch image and strict antitonicity in `LowerLambertW.lean`, the global lower bound and the exhaustive declaration `eq_principalLambertW_or_eq_lowerLambertW` in `LambertBranchDichotomy.lean`, this formalizes the complete real existence, range, and two-solution classification of Theorem 2.1. The endpoint-inclusive principal restriction is also the range theorem used by the formal two-branch power–exponential inverse package below. The endpoint continuity now has theorem-level names as well: `principalLambertW_continuousWithinAt_branchPoint` and `principalLambertW_continuousOn_Ici` cover the principal branch at the branch point and on its full real domain, while `lowerLambertW_continuousWithinAt_branchPoint` and `lowerLambertW_continuousOn_Ico` do the same for the lower branch. These continuity declarations are inputs to the compiled one-sided derivative and leading square-root layers recorded in remark 9.2.

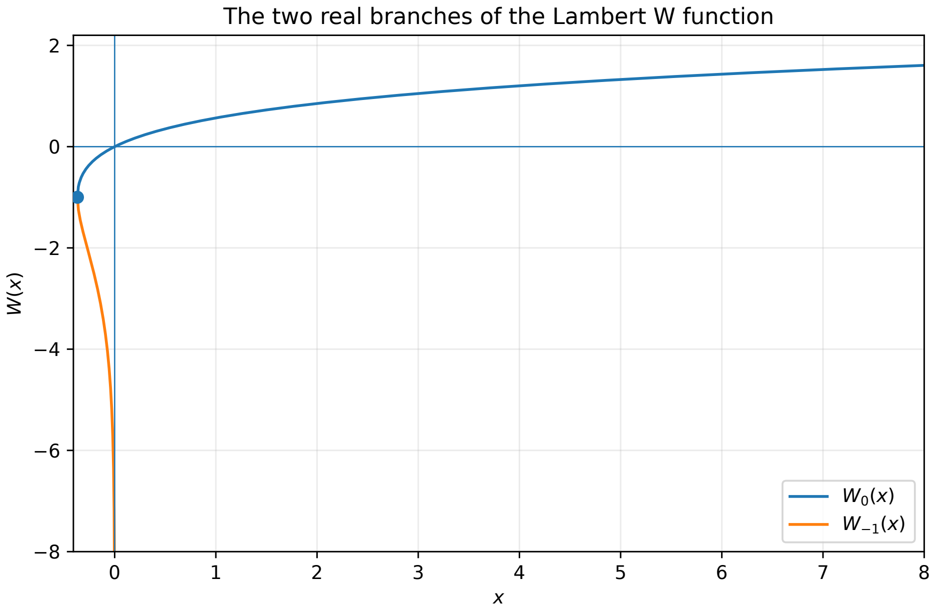

Figure 1: The principal branch W_0 and the nonprincipal real branch W_{-1}. They meet at the square-root branch point $(-1/\mathrm{e}, -1)$.

Branch warning 2.3 (The subscript matters). For $-1/\mathrm{e} < x < 0$, the notation $W(x)$ is incomplete unless the branch is understood. For example,

$$W_0(-0.1) \approx -0.1118326, \qquad W_{-1}(-0.1) \approx -3.5771521.$$

Both numbers satisfy $w\mathrm{e}^w = -0.1$.

Corollary 2.4 (Branch-aware inverse identities). *For real a,*

$$W_0(a\mathrm{e}^a) = a \quad \textit{when } a \geq -1,$$

and

$$W_{-1}(a\mathrm{e}^a) = a \quad \textit{when } a \leq -1.$$

At $a = -1$, both formulas hold.

Proof. The first statement says that W_0 undoes the restriction of $f(a) = a\mathrm{e}^a$ to $[-1, \infty)$; the second says the same about the restriction to $(-\infty, -1]$. This is precisely Theorem 2.1. □

Example 2.5 (A useful logarithmic identity). *Let $x > 0$. Taking $a = \log x$ in corollary 2.4 gives*

$$W_0(x \log x) = \log x \quad \textit{if } x \geq \mathrm{e}^{-1},$$

whereas

$$W_{-1}(x \log x) = \log x \quad \textit{if } 0 < x \leq \mathrm{e}^{-1}.$$

The same expression $x \log x$ therefore selects different inverse branches on opposite sides of $x = \mathrm{e}^{-1}$.

3 Elementary identities and transformations

Let $w = W_k(x)$ on any branch on which the displayed quantities are defined. The defining equation immediately yields

$$\mathrm{e}^w = \frac{x}{w}, \qquad \mathrm{e}^{-w} = \frac{w}{x}, \qquad \log|x| = w + \log|w| \tag{2}$$

for real $x \neq 0$. The last identity is often more stable numerically than $we^w = x$, because it avoids overflow and underflow.

Proposition 3.1 (An addition identity). *Suppose $a = W_j(x)$, $b = W_\ell(y)$, and $a, b, ab \neq 0$. Then*

$$(a+b)e^{a+b} = xy\left(\frac{1}{a} + \frac{1}{b}\right). \tag{3}$$

Consequently

$$a + b = W_m\left(xy\left(\frac{1}{a} + \frac{1}{b}\right)\right)$$

for whichever branch m contains the value $a + b$.

Proof. Since $e^a = x/a$ and $e^b = y/b$,

$$(a+b)e^{a+b} = (a+b)\frac{x}{a}\frac{y}{b} = xy\frac{a+b}{ab} = xy\left(\frac{1}{a} + \frac{1}{b}\right).$$

Applying the inverse branch containing $a + b$ proves the second statement. □

The final branch qualification is essential. Identities involving an inverse function cannot determine a branch merely by algebra.

3.1 The Wright omega viewpoint

For $t \in \mathbb{R}$, define

$$\omega(t) := W_0(e^t).$$

Then $\omega(t) > 0$ and

$$\omega(t) + \log \omega(t) = t. \tag{4}$$

Conversely, a positive solution of (4) is $\omega(t)$. This reparametrization is valuable when e^t is too large to store.

There is an equally natural two-branch negative analogue. For $u \geq 1$, the equation

$$v - \log v = u \tag{5}$$

has two positive solutions, one in $(0, 1]$ and one in $[1, \infty)$:

$$v_0(u) = -W_0(-e^{-u}), \qquad v_1(u) = -W_{-1}(-e^{-u}).$$

Proof. Writing $w = -v$ in $we^w = -e^{-u}$ gives $ve^{-v} = e^{-u}$. Taking logarithms gives $v - \log v = u$. The function $v - \log v$ decreases on $(0, 1]$, increases on $[1, \infty)$, and has minimum 1 at $v = 1$, so the two solutions and their locations follow. □

4 Exact relations between the two real branches

On the interval $(-1/e, 0)$, the two real branches are not independent. They are the two preimages of the same number under $w \mapsto we^w$. The following parametrization describes the pair exactly and, in particular, places Bernoulli numbers naturally into the theory.

Theorem 4.1 (Exact branch-pair parametrization). *Let $x \in (-1/e, 0)$, and define the positive branch gap*

$$\Delta = W_0(x) - W_{-1}(x) > 0. \tag{6}$$

Then

$$W_0(x) = -\frac{\Delta}{e^{\Delta} - 1}, \tag{7}$$

$$W_{-1}(x) = -\frac{\Delta}{1 - e^{-\Delta}} = -\frac{\Delta e^{\Delta}}{e^{\Delta} - 1}, \tag{8}$$

$$x = -\frac{\Delta}{e^{\Delta} - 1} \exp\left(-\frac{\Delta}{e^{\Delta} - 1}\right). \tag{9}$$

Conversely, every $\Delta > 0$ *inserted into these formulas produces one and only one* $x \in (-1/e, 0)$*, with the displayed values equal to* $W_0(x)$ *and* $W_{-1}(x)$*.*

Equivalently, if $t = e^{\Delta} > 1$*, then*

$$W_0(x) = -\frac{\log t}{t - 1}, \qquad W_{-1}(x) = -\frac{t \log t}{t - 1}, \tag{10}$$

$$x = -\frac{\log t}{t - 1}\, t^{-1/(t-1)}. \tag{11}$$

Proof. Put $a = W_0(x)$ and $b = W_{-1}(x)$. Then $a - b = \Delta$, and the equality $ae^a = be^b = x$ gives

$$\frac{a}{b} = e^{b-a} = e^{-\Delta}.$$

Thus $a = be^{-\Delta}$, and subtraction yields

$$\Delta = a - b = b(e^{-\Delta} - 1).$$

Solving first for b, and then for a, proves (7) and (8). Substitution into $x = ae^a$ proves (9). Replacing e^{Δ} by t gives (10) and (11).

It remains to justify the converse and the stated range of x. For $\Delta > 0$, the elementary inequality $e^{\Delta} > 1 + \Delta$ gives

$$-1 < -\frac{\Delta}{e^{\Delta} - 1} < 0.$$

Also

$$-\frac{\Delta}{1 - e^{-\Delta}} < -1,$$

because $1 - e^{-\Delta} < \Delta$. The two displayed numbers have positive gap Δ, and direct substitution shows that their products with their exponentials are equal. Their common value is therefore in $(-1/e, 0)$ by Theorem 2.1, and the branch ranges identify them uniquely as W_0 and W_{-1}. This also proves that every $\Delta > 0$ gives exactly one paired input. □

Corollary 4.2 (Exact symmetric identities). *For* $-1/e < x < 0$*, with* Δ *as in* (6)*,*

$$\frac{W_{-1}(x)}{W_0(x)} = e^{\Delta}, \tag{12}$$

$$W_0(x) + W_{-1}(x) = -\Delta \coth\left(\frac{\Delta}{2}\right), \tag{13}$$

$$W_0(x)W_{-1}(x) = \frac{\Delta^2}{4\sinh^2(\Delta/2)}. \tag{14}$$

In particular,

$$W_0(x) + W_{-1}(x) < -2, \qquad 0 < W_0(x)W_{-1}(x) < 1. \tag{15}$$

Proof. The ratio follows immediately from (7)–(8). Adding and multiplying those formulas, and using

$$\coth(y) = \frac{e^{2y}+1}{e^{2y}-1}, \qquad 4\sinh^2(y) = e^{2y} - 2 + e^{-2y},$$

gives (13) and (14).

For $y > 0$, $\sinh y > y$, so the product in (14) lies strictly between 0 and 1. Also $y\cosh y - \sinh y$ vanishes at 0 and has derivative $y \sinh y > 0$; hence $y \coth y > 1$. Taking $y = \Delta/2$ proves the sum inequality. ☐

4.1 Bernoulli-number expansions of the paired branches

The Bernoulli numbers B_n are defined by the generating function

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}, \qquad |z| < 2\pi. \tag{16}$$

At $z = 0$, the left side of (16) is understood in the standard removable-origin sense, with value 1. The Lean crosswalk below represents that convention by $\texttt{complexExpm1Div}(z)^{-1}$: this is 1 at the origin and is the literal quotient $z/(\exp z - 1)$ when $z \neq 0$. It does not identify the series with Lean's totalized literal quotient at zero, whose value is 0. Thus $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_4 = -1/30$, and all odd B_n beyond B_1 vanish.

Corollary 4.3 (Bernoulli expansion in the branch gap)**.** *For* $0 < \Delta < 2\pi$*,*

$$W_0(x) = -1 + \frac{\Delta}{2} - \frac{\Delta^2}{12} + \frac{\Delta^4}{720} - \frac{\Delta^6}{30240} + \cdots, \tag{17}$$

$$W_{-1}(x) = -1 - \frac{\Delta}{2} - \frac{\Delta^2}{12} + \frac{\Delta^4}{720} - \frac{\Delta^6}{30240} + \cdots. \tag{18}$$

More compactly,

$$W_0(x) = -\sum_{n=0}^{\infty} B_n \frac{\Delta^n}{n!}, \qquad W_{-1}(x) = -\Delta - \sum_{n=0}^{\infty} B_n \frac{\Delta^n}{n!}. \tag{19}$$

Proof. The first formula in (19) is (7) together with (16). The identity

$$\frac{\Delta}{1 - e^{-\Delta}} = \Delta + \frac{\Delta}{e^{\Delta} - 1}$$

gives the second. Substitution of the first Bernoulli numbers gives the expanded forms. The convergence radius is 2π, because the nearest nonzero zeros of $e^z - 1$ are $z = \pm 2\pi i$. ☐

Remark 4.4. The gap variable Δ is different from the square-root variable used near the branch point in Section 9. They are asymptotically equivalent up to a factor: by (54), $\Delta \sim 2\sqrt{2(1 + ex)}$. The square-root variable is best for evaluating a single branch near $-1/e$; the gap variable is best for relations that involve both branches at once.

5 Differential calculus and geometry

5.1 First and second derivatives

Theorem 5.1 (Derivative formulas)**.** *For the principal branch assume* $x > -1/e$*, and for the lower real branch assume* $-1/e < x < 0$*. On these respective smooth real domains,*

$$W_k'(x) = \frac{1}{e^{W_k(x)}(1 + W_k(x))}, \tag{20}$$

$$W_k''(x) = -\frac{e^{-2W_k(x)}(W_k(x)+2)}{(1+W_k(x))^3}. \tag{21}$$

If in addition $x \neq 0$, these may be rewritten as

$$W_k'(x) = \frac{W_k(x)}{x(1+W_k(x))}, \tag{22}$$

$$W_k''(x) = -\frac{W_k(x)^2(W_k(x)+2)}{x^2(1+W_k(x))^3}. \tag{23}$$

At the regular principal-branch point $x = 0$, stated separately rather than through an expression with a zero denominator,

$$W_0'(0) = 1, \qquad W_0''(0) = -2.$$

Proof. Write $w = W_k(x)$. Differentiating $we^w = x$ gives

$$e^w(1+w)w' = 1,$$

which proves (20). Since $e^{-w} = w/x$ when $x \neq 0$, this also proves (22).

Now regard w' as a function of w:

$$w' = \frac{e^{-w}}{1+w}.$$

Differentiate with respect to w, then multiply by w':

$$w'' = \left(-\frac{e^{-w}}{1+w} - \frac{e^{-w}}{(1+w)^2}\right)\frac{e^{-w}}{1+w} = -\frac{e^{-2w}(w+2)}{(1+w)^3}.$$

When $x \neq 0$, substituting $e^{-w} = w/x$ proves (23). At $x = 0$ on the principal branch, $w = 0$, so the exponential-denominator formulas give the two stated values directly; they also follow from the power series proved in Section 8. □

Corollary 5.2 (Monotonicity recovered from the derivative)**.** *On $(-1/e, \infty)$, $W_0'(x) > 0$. On $(-1/e, 0)$, $W_{-1}'(x) < 0$. Moreover,*

$$\lim_{x \downarrow -1/e} W_0'(x) = +\infty, \qquad \lim_{x \downarrow -1/e} W_{-1}'(x) = -\infty.$$

Proof. Use (20) and the ranges in Theorem 2.1. At the common endpoint, $1 + W_k(x) \to 0$, with positive sign on the principal branch and negative sign on the nonprincipal branch. The remaining factors tend to nonzero finite values. □

Remark 5.3 (Formalized real-branch curvature). The first-derivative identities and their strict signs on the two branch interiors are formalized in `PrincipalLambertW.lean` and `LowerLambertW.lean`; the scaled power–exponential specializations are in `PowerExponentialLambertCalculus.lean`. The endpoint identity $W_0'(0) = 1$ is `deriv_principalLambertW_zero`.

The complete raw second-order package is now `LambertWCurvature.lean`. On the open principal domain, `deriv_principalLambertW`, `deriv_principalLambertW_hasDerivAt`, and `deriv_deriv_principalLambertW` give the nonsingular exponential forms in (20) and (21); `deriv_deriv_principalLambertW_zero` gives $W_0''(0) = -2$, and `deriv_deriv_principalLambertW_neg` gives strict negativity. The theorem `strictConcaveOn_principalLambertW` strengthens the open interval statement below to strict concavity on the full closed domain $[-1/e, \infty)$.

For the lower branch, `deriv_lowerLambertW_hasDerivAt` and `deriv_deriv_lowerLambertW` give the exact second derivative on $(-1/e, 0)$. The declarations `deriv_deriv_lowerLambertW_pos_iff`, `deriv_deriv_lowerLambertW_neg_iff`, and `deriv_deriv_` `lowerLambertW_eq_zero_iff` give its complete sign and unique-zero classification at $-2/\mathrm{e}^2$. Finally, `strictConvexOn_lowerLambertW_left` proves strict convexity on $[-1/\mathrm{e}, -2/\mathrm{e}^2]$, and `strictConcaveOn_lowerLambertW_right` proves strict concavity on $[-2/\mathrm{e}^2, 0)$.

The raw one-sided endpoint limits, their secant forms, and the resulting failure of finite differentiability are now packaged in `LambertWBranchPointGeometry.lean`; the exact declaration crosswalk and the boundary between the leading equivalent and the stronger Puiseux theorem appear in remark 9.2.

Theorem 5.4 (Concavity and the inflection point)**.** *The principal branch W_0 is strictly concave on $(-1/\mathrm{e}, \infty)$. The branch W_{-1} is*

- *strictly convex on $(-1/\mathrm{e}, -2/\mathrm{e}^2)$,*
- *has an inflection point at $x = -2/\mathrm{e}^2$, where $W_{-1}(x) = -2$,*
- *strictly concave on $(-2/\mathrm{e}^2, 0)$.*

Proof. For W_0, one has $w > -1$, hence $w + 2 > 0$ and $(1 + w)^3 > 0$. Formula (21) therefore gives $w'' < 0$.

For W_{-1}, one has $w < -1$, so $(1 + w)^3 < 0$. If $-2 < w < -1$, then $w + 2 > 0$, and (21) gives $w'' > 0$. If $w < -2$, then $w + 2 < 0$, and $w'' < 0$. The value $w = -2$ corresponds to $x = w\mathrm{e}^w = -2/\mathrm{e}^2$. The branch is decreasing, so the stated x-intervals follow. ☐

5.2 Higher derivatives

Repeated differentiation produces a structured family of polynomials.

Theorem 5.5 (Polynomial recurrence for all derivatives)**.** *Define polynomials P_n by*

$$P_1(w) = 1, \qquad P_{n+1}(w) = (1 + w)P_n'(w) - (nw + 3n - 1)P_n(w).$$

Then, for every $n \geq 1$,

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} W_k(x) = \frac{\mathrm{e}^{-nW_k(x)} P_n(W_k(x))}{(1 + W_k(x))^{2n-1}}. \tag{24}$$

The first polynomials are

$$P_1 = 1, \qquad P_2 = -(w + 2), \qquad P_3 = 2w^2 + 8w + 9,$$
$$P_4 = -(6w^3 + 36w^2 + 79w + 64).$$

Proof. The formula for $n = 1$ is (20). Assume (24) holds for n, and differentiate it first with respect to w:

$$\frac{\mathrm{d}}{\mathrm{d}w}\left(\frac{\mathrm{e}^{-nw} P_n(w)}{(1 + w)^{2n-1}}\right) = \frac{\mathrm{e}^{-nw}}{(1 + w)^{2n}} \left((1 + w)P_n'(w) - (nw + 3n - 1)P_n(w)\right).$$

Multiplication by $\mathrm{d}w/\mathrm{d}x = \mathrm{e}^{-w}/(1 + w)$ yields

$$\frac{\mathrm{e}^{-(n+1)w} P_{n+1}(w)}{(1 + w)^{2n+1}},$$

which is (24) with $n + 1$. The displayed polynomials follow by direct substitution into the recurrence. ☐

5.3 Conditioning

Suppose x is perturbed by a small amount Δx. Linearization suggests

$$\Delta W \approx W'(x)\Delta x.$$

For nonzero x and W, the relative condition number is

$$\kappa_{\mathrm{rel}}(x) = \left|\frac{xW'(x)}{W(x)}\right| = \frac{1}{|1+W(x)|}. \tag{25}$$

Proof. Insert (22) into the definition of relative condition number. □

Thus both real branches are badly conditioned near $-1/\mathrm{e}$, where $W=-1$. By contrast, $\kappa_{\mathrm{rel}} \to 0$ both as $x \to +\infty$ on W_0 and as $x \to 0^-$ on W_{-1}. This distinction is important: the branch point is intrinsically sensitive, while the logarithmic singularity of W_{-1} at zero is comparatively well conditioned in a relative sense.

6 Integral calculus and exact moments

The substitution

$$w = W_k(x), \qquad x = w\mathrm{e}^w, \qquad \mathrm{d}x = \mathrm{e}^w(1+w)\,\mathrm{d}w$$

turns many apparently non-elementary integrals into an exponential times a polynomial.

Theorem 6.1 (Two basic antiderivatives)**.** *On any interval contained in a differentiable branch,*

$$\int W_k(x)\,\mathrm{d}x = \mathrm{e}^{W_k(x)}\left(W_k(x)^2 - W_k(x) + 1\right) + C \tag{26}$$

$$= x\left(W_k(x) - 1 + \frac{1}{W_k(x)}\right) + C, \tag{27}$$

where the second form is used away from $x=0$. Also, away from $x=0$,

$$\int \frac{W_k(x)}{x}\,\mathrm{d}x = W_k(x) + \frac{1}{2}W_k(x)^2 + C. \tag{28}$$

Proof. With $x = w\mathrm{e}^w$,

$$\int W_k(x)\,\mathrm{d}x = \int w\mathrm{e}^w(1+w)\,\mathrm{d}w = \int (w+w^2)\mathrm{e}^w\,\mathrm{d}w.$$

Direct differentiation shows that

$$\frac{\mathrm{d}}{\mathrm{d}w}\left[\mathrm{e}^w(w^2-w+1)\right] = \mathrm{e}^w(w+w^2),$$

which proves (26). Since $\mathrm{e}^w = x/w$, it also proves (27). Furthermore,

$$\frac{\mathrm{d}x}{x} = \frac{1+w}{w}\,\mathrm{d}w,$$

and therefore

$$\int \frac{W_k(x)}{x}\,\mathrm{d}x = \int w\frac{1+w}{w}\,\mathrm{d}w = \int (1+w)\,\mathrm{d}w = w + \frac{1}{2}w^2 + C.$$

□

Corollary 6.2 (Two branch-dependent definite integrals). *The two real branches give*

$$\int_{-1/\mathrm{e}}^{0} W_0(x)\,\mathrm{d}x = 1 - \frac{3}{\mathrm{e}}, \tag{29}$$

$$\int_{-1/\mathrm{e}}^{0} W_{-1}(x)\,\mathrm{d}x = -\frac{3}{\mathrm{e}}. \tag{30}$$

Both integrals are improper only in the harmless sense that $W_{-1}(x) \to -\infty$ at 0; the second integral nevertheless converges.

Proof. Use the antiderivative $F(w) = \mathrm{e}^w(w^2 - w + 1)$. On the principal branch, w runs from -1 to 0, so

$$F(0) - F(-1) = 1 - \frac{3}{\mathrm{e}}.$$

On the nonprincipal branch, w runs from -1 to $-\infty$. Since e^w decreases faster than every polynomial grows, $\lim_{w\to-\infty} F(w) = 0$. Hence the integral is $0 - 3/\mathrm{e}$. ☐

6.1 A finite formula for polynomial moments

For $a \neq 0$ and an integer $q \geq 0$, define

$$Q_q^{(a)}(w) = \sum_{j=0}^{q} \frac{(-1)^j q!}{(q-j)!\,a^{j+1}} w^{q-j}. \tag{31}$$

Lemma 6.3. *For $a \neq 0$ and $q \geq 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}w}\left(\mathrm{e}^{aw} Q_q^{(a)}(w)\right) = \mathrm{e}^{aw} w^q.$$

Proof. Differentiate the finite sum. The term $aQ_q^{(a)}$ contributes

$$\sum_{j=0}^{q} \frac{(-1)^j q!}{(q-j)!\,a^{j}} w^{q-j}.$$

The derivative $Q_q^{(a)\prime}$ contributes

$$\sum_{j=0}^{q-1} \frac{(-1)^j q!}{(q-j-1)!\,a^{j+1}} w^{q-j-1}.$$

Replace j by $j-1$ in the second sum. Every term except w^q cancels with the corresponding term of the first sum. ☐

Theorem 6.4 (All nonnegative integral moments). *Let m, n be nonnegative integers, put $a = m+1$ and $q = m+n$. Then*

$$\int x^m W_k(x)^n\,\mathrm{d}x = \mathrm{e}^{aW_k(x)}\left(Q_q^{(a)}(W_k(x)) + Q_{q+1}^{(a)}(W_k(x))\right) + C. \tag{32}$$

Thus every integral of a nonnegative integer power of x times a nonnegative integer power of $W_k(x)$ has an elementary expression in $W_k(x)$.

Proof. With $x = w\mathrm{e}^w$,

$$x^m W_k(x)^n\,\mathrm{d}x = w^{m+n}\mathrm{e}^{mw}\mathrm{e}^w(1+w)\,\mathrm{d}w = \mathrm{e}^{aw}(w^q + w^{q+1})\,\mathrm{d}w.$$

Apply lemma 6.3 to the two terms. ☐

Remark 6.5. The same method works when the exponent of x is an integer less than -1, provided the corresponding powers of w can be integrated. The special case $m = -1$ is simpler and leads to (28). Fractional powers require a consistent choice of the real or complex power and generally lead to incomplete gamma functions.

6.2 Mellin-type definite integrals on the unbounded ends

The same substitution also evaluates a continuous family of improper integrals. Recall

$$\Gamma(s) = \int_0^\infty t^{s-1} \mathrm{e}^{-t}\,\mathrm{d}t \quad (s > 0),$$

and define the upper incomplete gamma integral by

$$\Gamma(s, c) = \int_c^\infty t^{s-1} \mathrm{e}^{-t}\,\mathrm{d}t \quad (c > 0).$$

Because the latter integral starts at a positive number, it converges for every real s.

Theorem 6.6 (Mellin integrals for both unbounded real ends)**.** *Let $\alpha, \beta \in \mathbb{R}$.*

(i) *If $\beta > 1$ and $\alpha - \beta > -1$, then*

$$\int_0^\infty \frac{W_0(x)^\alpha}{x^\beta}\,\mathrm{d}x = \frac{\Gamma(\alpha-\beta+1)}{(\beta-1)^{\alpha-\beta+1}} + \frac{\Gamma(\alpha-\beta+2)}{(\beta-1)^{\alpha-\beta+2}} \tag{33}$$

$$= \frac{\alpha\,\Gamma(\alpha-\beta+1)}{(\beta-1)^{\alpha-\beta+2}}. \tag{34}$$

(ii) *If $\beta > -1$, then*

$$\int_{-1/\mathrm{e}}^0 [-W_{-1}(x)]^\alpha (-x)^\beta\,\mathrm{d}x$$
$$= \frac{\Gamma(\alpha+\beta+2, \beta+1)}{(\beta+1)^{\alpha+\beta+2}} - \frac{\Gamma(\alpha+\beta+1, \beta+1)}{(\beta+1)^{\alpha+\beta+1}}. \tag{35}$$

Equivalently, using the incomplete-gamma recurrence,

$$\int_{-1/\mathrm{e}}^0 [-W_{-1}(x)]^\alpha (-x)^\beta\,\mathrm{d}x = \frac{\alpha\,\Gamma(\alpha+\beta+1, \beta+1)}{(\beta+1)^{\alpha+\beta+2}} + \frac{\mathrm{e}^{-(\beta+1)}}{\beta+1}. \tag{36}$$

Proof. For part (i), put $w = W_0(x)$. As x runs from 0 to ∞, w runs from 0 to ∞, and

$$x = w\mathrm{e}^w, \qquad \mathrm{d}x = \mathrm{e}^w(1+w)\,\mathrm{d}w.$$

Consequently, with $c = \beta - 1 > 0$ and $\nu = \alpha - \beta > -1$,

$$\int_0^\infty \frac{W_0(x)^\alpha}{x^\beta}\,\mathrm{d}x = \int_0^\infty w^\nu (1+w)\mathrm{e}^{-cw}\,\mathrm{d}w$$
$$= \frac{\Gamma(\nu+1)}{c^{\nu+1}} + \frac{\Gamma(\nu+2)}{c^{\nu+2}}.$$

This proves (33). Since $\Gamma(\nu+2) = (\nu+1)\Gamma(\nu+1)$, the two terms combine to (34). The hypotheses are also exactly what is needed for convergence at $w = 0$ and at $w = \infty$.

For part (ii), put $y = -W_{-1}(x)$. Then y runs from 1 to ∞,

$$x = -y\mathrm{e}^{-y}, \qquad \mathrm{d}x = (y-1)\mathrm{e}^{-y}\,\mathrm{d}y.$$

Writing $p = \alpha + \beta$ and $c = \beta + 1 > 0$, we obtain

$$\int_{-1/\mathrm{e}}^0 [-W_{-1}(x)]^\alpha (-x)^\beta\,\mathrm{d}x = \int_1^\infty y^p (y-1)\mathrm{e}^{-cy}\,\mathrm{d}y$$
$$= \int_1^\infty \left(y^{p+1} - y^p\right) \mathrm{e}^{-cy}\,\mathrm{d}y.$$

In each term let $t = cy$. This gives (35). Finally, integration by parts proves

$$\Gamma(s+1, c) = s\Gamma(s, c) + c^s \mathrm{e}^{-c},$$

and substitution with $s = \alpha + \beta + 1$ gives (36). □

Example 6.7. *Taking* $(\alpha, \beta) = (1, 3/2)$ *in* (34) *gives the attractive identity*

$$\int_0^\infty \frac{W_0(x)}{x\sqrt{x}}\,\mathrm{d}x = 2\sqrt{2\pi}. \tag{37}$$

It is an exact evaluation of an integral whose integrand contains a non-elementary inverse function.

7 Taylor series at an arbitrary regular point

The derivative recurrence of Theorem 5.5 immediately gives a local series about any point away from the branch point.

Theorem 7.1 (Local Taylor expansion). *Let* x_0 *be a point on a branch* W_k, *and let* $w_0 = W_k(x_0) \neq -1$. *In a sufficiently small neighborhood of* x_0,

$$W_k(x) = w_0 + \sum_{n=1}^{\infty} \frac{\mathrm{e}^{-nw_0} P_n(w_0)}{n!(1+w_0)^{2n-1}}(x - x_0)^n, \tag{38}$$

where the polynomials P_n *are those of Theorem 5.5. This formula applies without change to* W_0 *and* W_{-1}, *provided* x_0 *belongs to the selected branch.*

Proof. The inverse function theorem gives a differentiable, in fact analytic, local inverse of $w\mathrm{e}^w$ at w_0, because $\mathrm{e}^{w_0}(1 + w_0) \neq 0$. Taylor's formula therefore applies, and Theorem 5.5 supplies its coefficients. □

Example 7.2. *At* $x_0 = 1$, $w_0 = \Omega$. *The first two terms are*

$$W_0(1 + h) = \Omega + \frac{\mathrm{e}^{-\Omega}}{1 + \Omega}h - \frac{\mathrm{e}^{-2\Omega}(\Omega + 2)}{2(1 + \Omega)^3}h^2 + \mathcal{O}(h^3).$$

Since $\mathrm{e}^{-\Omega} = \Omega$, *every coefficient can be written as a rational function of* Ω.

The singular point $w_0 = -1$ is deliberately excluded. There the inverse has a square-root, not an ordinary Taylor, expansion; this is the subject of Section 9.

8 The Maclaurin series of the principal branch

The principal branch is analytic at the origin. Its coefficients have an unexpectedly simple closed form.

For a formal power series $A(t)$, the notation $[t^n]A(t)$ means the coefficient of t^n. We use the following form of Lagrange inversion, proved in Appendix A.

Theorem 8.1 (Lagrange–Buermann inversion). *Suppose* $y = x\phi(y)$, *where* $\phi(0) \neq 0$, *and let* H *be a formal power series. For* $n \geq 1$,

$$[x^n]H(y(x)) = \frac{1}{n}[t^{n-1}]H'(t)\phi(t)^n. \tag{39}$$

In particular, for integers $m \geq 1$ *and* $n \geq m$,

$$[x^n]y(x)^m = \frac{m}{n}[t^{n-m}]\phi(t)^n. \tag{40}$$

The defining equation for W can be written

$$W = x\mathrm{e}^{-W}.$$

Thus $\phi(t) = \mathrm{e}^{-t}$.

Theorem 8.2 (Maclaurin series and powers). *For* $|x| < 1/\mathrm{e}$,

$$\boxed{W_0(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} x^n} = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + \frac{125}{24}x^5 - \cdots . \tag{41}$$

More generally, for every integer $m \geq 1$,

$$W_0(x)^m = \sum_{n=m}^{\infty} \frac{m}{n} \frac{(-n)^{n-m}}{(n-m)!} x^n. \tag{42}$$

Consequently

$$W_0^{(n)}(0) = (-n)^{n-1} \qquad (n \geq 1). \tag{43}$$

The radius of convergence is exactly $1/\mathrm{e}$.

Proof. Apply (40) with $\phi(t) = \mathrm{e}^{-t}$:

$$[x^n] W(x)^m = \frac{m}{n} [t^{n-m}] \mathrm{e}^{-nt} = \frac{m}{n} \frac{(-n)^{n-m}}{(n-m)!}.$$

This proves (42), and $m = 1$ gives (41). The derivative formula follows by multiplying the coefficient of x^n by $n!$.

Let

$$a_n = \frac{n^{n-1}}{n!}.$$

Then

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n}\right)^{n-1} \longrightarrow \mathrm{e}.$$

The ratio test therefore gives radius $1/\mathrm{e}$. Within that disk, the series produced by Lagrange inversion satisfies $W = x\mathrm{e}^{-W}$ and has $W(0) = 0$. The local inverse is unique, so it is the principal branch. □

Remark 8.3 (What the radius means). The nearest singularity to the origin is the branch point $-1/\mathrm{e}$. The coefficient calculation detects exactly that distance without requiring us to know the complex geometry in advance.

8.1 Rigorous remainder estimates

Let

$$T_n(x) = \frac{(-n)^{n-1}}{n!} x^n.$$

A direct calculation gives

$$\left| \frac{T_{n+1}(x)}{T_n(x)} \right| = |x| \left(1 + \frac{1}{n}\right)^{n-1} < \mathrm{e}\,|x| \,. \tag{44}$$

Theorem 8.4 (Geometric tail bound). *If* $|x| < 1/\mathrm{e}$, $q = \mathrm{e}\,|x| < 1$, *and* $S_N(x) = \sum_{n=1}^{N} T_n(x)$, *then*

$$|W_0(x) - S_N(x)| \leq \frac{|T_{N+1}(x)|}{1-q}. \tag{45}$$

If $0 \leq x \leq 1/\mathrm{e}$, *the terms alternate with decreasing magnitudes, so the sharper estimate*

$$|W_0(x) - S_N(x)| \leq |T_{N+1}(x)| \tag{46}$$

holds, and the error has the sign of $T_{N+1}(x)$.

Proof. Equation (44) shows that every successive magnitude in the tail is at most q times the preceding magnitude. Summing the resulting geometric majorant proves (45). For $x \geq 0$, the factor $(-n)^{n-1}$ makes the signs alternate. The same ratio is at most 1 on $0 \leq x \leq 1/\mathrm{e}$, so the alternating-series theorem proves (46). □

For negative x, every nonzero term in (41) is negative. At the endpoint $x = -1/\mathrm{e}$, the series still converges absolutely. One way to see this without invoking the full Stirling formula is Raabe's test: if

$$b_n = \frac{n^{n-1}}{n!\mathrm{e}^n},$$

then

$$\log \frac{b_n}{b_{n+1}} = 1 - (n-1)\log\left(1 + \frac{1}{n}\right) = \frac{3}{2n} + \mathcal{O}(n^{-2}).$$

Hence $b_n/b_{n+1} = 1 + 3/(2n) + \mathcal{O}(n^{-2})$, and Raabe's test applies because $3/2 > 1$. Moreover, for every $x \in [-1/\mathrm{e}, 0]$, the magnitude of the n-th term is at most b_n. The Weierstrass M-test therefore gives uniform convergence on this closed interval. The sum is continuous there, agrees with W_0 on $(-1/\mathrm{e}, 0]$, and hence agrees at the left endpoint by continuity. Consequently

$$\sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} \left(-\frac{1}{\mathrm{e}}\right)^n = W_0(-1/\mathrm{e}) = -1.$$

The one-sided branch-point continuity used in this endpoint argument is now available as `principalLambertW_continuousWithinAt_branchPoint`; that declaration does not itself formalize the Maclaurin series or its endpoint sum.

8.2 The tree function and Cayley's formula

Define

$$T(x) := -W_0(-x). \tag{47}$$

Then

$$T(x) = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} x^n, \qquad T = x\mathrm{e}^T.$$

The coefficient n^{n-1} counts rooted labelled trees on n vertices.

Theorem 8.5 (Cayley and the tree coefficients)**.** *There are n^{n-2} unrooted trees on the labelled vertex set $\{1, \ldots, n\}$, and n^{n-1} rooted labelled trees. Therefore the exponential generating function for rooted labelled trees is exactly $T(x) = -W_0(-x)$.*

Proof. The Prufer code of a labelled tree is formed by repeatedly deleting the smallest-labelled leaf and recording its neighbor, until two vertices remain. This produces a sequence of length $n - 2$ with entries in $\{1, \ldots, n\}$.

Conversely, given such a sequence, repeatedly join its first entry to the smallest label not appearing in the remaining sequence, delete that first entry, and finally join the last two unused labels. These two procedures are inverse to each other. Thus labelled trees are in bijection with the n^{n-2} sequences of length $n - 2$. Choosing one of the n vertices as root gives $n \cdot n^{n-2} = n^{n-1}$ rooted trees.

It remains only to compare coefficients: (47) and (41) give $n^{n-1}/n!$, the exponential-generating-function coefficient for n^{n-1} labelled objects. □

There is also a structural explanation of $T = x\mathrm{e}^T$. A rooted labelled tree consists of a root together with an unordered set of rooted labelled trees attached to it. In the language of exponential generating functions, an unordered labelled set contributes an exponential, giving $T = x\exp(T)$. Lagrange inversion then recovers Cayley's coefficient.

9 The square-root expansion at the branch point

Ordinary Taylor series fail at $x = -1/\mathrm{e}$: from the right the principal derivative diverges to $+\infty$, the lower derivative diverges to $-\infty$, and neither branch has a finite right derivative there. The correct local variable is a square root.

Put

$$w = -1 + s.$$

Then

$$\mathrm{e}x = (-1+s)\mathrm{e}^s,$$

and expansion at $s = 0$ gives

$$2(1+\mathrm{e}x) = \sum_{m=2}^{\infty} \frac{2(m-1)}{m!} s^m = s^2 + \frac{2}{3}s^3 + \frac{1}{4}s^4 + \frac{1}{15}s^5 + \cdots . \tag{48}$$

Define the *signed* square-root coordinates

$$p_0(x) = +\sqrt{2(1+\mathrm{e}x)}, \qquad p_{-1}(x) = -\sqrt{2(1+\mathrm{e}x)}. \tag{49}$$

Theorem 9.1 (Unified Puiseux expansion). *There is a convergent power series $S(p) = \sum_{n\geq 1} a_n p^n$ near $p = 0$ such that*

$$W_k(x) = -1 + S(p_k(x)), \qquad k \in \{0, -1\}. \tag{50}$$

Its first terms are

$$S(p) = p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 + \mathcal{O}(p^7). \tag{51}$$

Thus the same series describes both real branches: only the sign of p changes.

Proof. For $s \neq 0$, divide (48) by s^2 and choose the analytic square root that equals 1 at $s = 0$:

$$p = s\sqrt{1 + \frac{2}{3}s + \frac{1}{4}s^2 + \cdots}.$$

This defines an analytic function $p = p(s)$ with $p(0) = 0$ and $p'(0) = 1$. The inverse function theorem gives an analytic inverse $s = S(p)$ near 0, with $S(0) = 0$ and $S'(0) = 1$. For the principal branch $s \geq 0$, so $p = p_0 \geq 0$. For the nonprincipal branch $s \leq 0$, so the analytic signed coordinate is $p = p_{-1} \leq 0$. Hence (50) holds.

To compute coefficients, substitute $S(p) = a_1 p + a_2 p^2 + \cdots$ into (48) and compare successive powers of p. This gives $a_1 = 1$, then

$$a_2 = -\frac{1}{3}, \quad a_3 = \frac{11}{72}, \quad a_4 = -\frac{43}{540}, \quad a_5 = \frac{769}{17280}, \quad a_6 = -\frac{221}{8505}.$$

Because each new comparison is linear in the next unknown coefficient, the procedure determines all coefficients uniquely. □

Here is an explicit recurrence that makes the final sentence of the proof algorithmic. Write

$$S(p) = \sum_{n\geq 1} a_n p^n, \qquad \mathrm{e}^{-S(p)} = \sum_{n\geq 0} b_n p^n.$$

Differentiating the defining relation gives

$$S(p)S'(p) = p\mathrm{e}^{-S(p)}.$$

Consequently

$$b_0 = 1, \qquad b_n = -\frac{1}{n}\sum_{j=1}^{n} j a_j b_{n-j}, \tag{52}$$

$$a_1 = 1, \qquad a_n = \frac{b_{n-1} - \sum_{j=2}^{n-1}(n-j+1)a_j a_{n-j+1}}{n+1} \quad (n \geq 2). \tag{53}$$

Proof. The identity $B' = -S'B$, where $B = \mathrm{e}^{-S}$, gives (52) by comparing the coefficient of p^{n-1}. The coefficient of p^n in $SS' = pB$ is

$$\sum_{j=1}^{n}(n-j+1)a_j a_{n-j+1} = b_{n-1}.$$

The terms $j = 1$ and $j = n$ together equal $(n+1)a_n$; moving the remaining terms to the other side gives (53). □

9.1 Even and odd combinations of the two branches

Let

$$q = \sqrt{2(1+\mathrm{e}x)} \geq 0.$$

Then $p_0 = q$ and $p_{-1} = -q$. Therefore

$$W_0(x) - W_{-1}(x) = 2q + \frac{11}{36}q^3 + \frac{769}{8640}q^5 + \mathcal{O}(q^7), \tag{54}$$

$$\frac{W_0(x) + W_{-1}(x)}{2} = -1 - \frac{1}{3}q^2 - \frac{43}{540}q^4 - \frac{221}{8505}q^6 + \mathcal{O}(q^8). \tag{55}$$

The branch difference contains only odd powers, while the branch mean contains only even powers. These formulas are immediate by adding and subtracting $S(q)$ and $S(-q)$.

The leading term gives the especially useful equivalence

$$W_{0,-1}(x) = -1 \pm \sqrt{2\mathrm{e}\left(x + \frac{1}{\mathrm{e}}\right)} + \mathcal{O}\left(x + \frac{1}{\mathrm{e}}\right), \tag{56}$$

where the plus sign is for W_0. It also explains the vertical tangents:

$$W_0'(x) \sim \frac{\mathrm{e}}{q}, \qquad W_{-1}'(x) \sim -\frac{\mathrm{e}}{q}.$$

Remark 9.2 (Compiled Lean branch-point crosswalk). The raw geometric endpoint module `LambertWBranchPointGeometry.lean` exposes exactly eight declarations. The one-sided derivative limits are `tendsto_deriv_principalLambertW_branchPoint_atTop` and `tendsto_deriv_lowerLambertW_branchPoint_atBot`; their endpoint-secant counterparts are `tendsto_principalLambertW_secantSlope_branchPoint_atTop` and `tendsto_lowerLambertW_secantSlope_branchPoint_atBot`. Failure of a finite right derivative is recorded by `principalLambertW_not_differentiableWithinAt_branchPoint` and `lowerLambertW_not_differentiableWithinAt_branchPoint`, while failure of ordinary differentiability of the totalized functions is `principalLambertW_not_differentiableAt_branchPoint` and `lowerLambertW_not_differentiableAt_branchPoint`.

The companion `LambertWBranchPointAsymptotics.lean` has one definition and eight theorems. The definition `lambertWBranchPointScale` is $q(x) = \sqrt{2\mathrm{e}(x + \mathrm{e}^{-1})}$, with `lamber` tWBranchPointScale_pos and lambertWBranchPointScale_sq recording its positivity and exact square. The two limits of the squared displacement divided by $x + \mathrm{e}^{-1}$ are tendsto_principalLambertW_add_one_sq_div_branchPoint and tendsto_lowerLambertW_add_one_sq_div_branchPoint; their asymptotic equivalence forms are principalLambertW_add_one_sq_isEquivalent_branchPoint and lowerLambertW_add_one_sq_isEquivalent_branchPoint. Finally, principalLambertW_add_one_isEquivalent_branchPoint and lowerLambertW_add_one_isEquivalent_branchPoint give precisely

$$W_0(x) + 1 \sim q(x), \qquad W_{-1}(x) + 1 \sim -q(x) \quad (x \downarrow -1/\mathrm{e}).$$

This formalizes the signed leading equivalent, but not the stronger $\mathcal{O}(x + \mathrm{e}^{-1})$ remainder in (56), any displayed higher coefficient in (51), the recurrences (52)–(53), or convergence of the complete series in Theorem 9.1. The derivative equivalents $W_k' \sim \pm\mathrm{e}/q$ shown above are also stronger than the compiled divergence statements. Finally, the generic power–exponential and classical Fabius phase wrappers, together with their peak-threshold vertical-tangent and square-root corollaries, remain unformalized; the raw two-branch results must still be transported through those parameter maps.

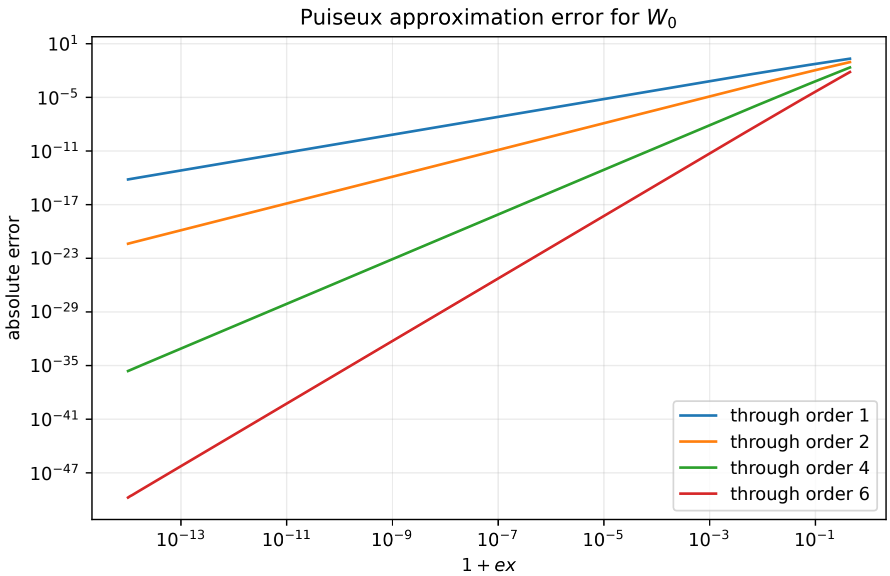

Figure 2: Absolute errors of successive truncations of (51) on the principal branch.

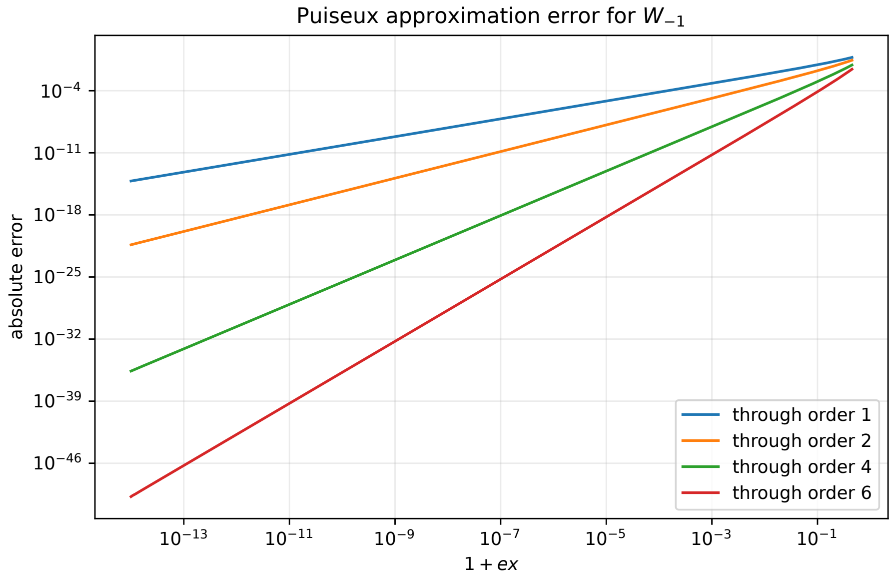

Figure 3: The same signed expansion, now with $p < 0$, approximates the nonprincipal branch.

9.2 An exponential branch-point coordinate

A second useful parametrization is

$$x = -\mathrm{e}^{-1-t^2/2}. \tag{57}$$

Choose $t > 0$ for W_0 and $t < 0$ for W_{-1}. If $W = -1 + u$, then the defining equation becomes

$$(1-u)\mathrm{e}^u = \mathrm{e}^{-t^2/2},$$

or

$$\frac{t^2}{2} = -u - \log(1-u). \tag{58}$$

The right side begins with $u^2/2$, so another inverse-series calculation gives

$$W = -1 + t - \frac{1}{3}t^2 + \frac{1}{36}t^3 + \frac{1}{270}t^4 + \frac{1}{4320}t^5 - \frac{1}{17010}t^6 + \mathcal{O}(t^7). \tag{59}$$

The proof is identical in structure to that of Theorem 9.1: the signed analytic square root of $2[-u - \log(1-u)]$ has derivative 1 at the origin and therefore has a local analytic inverse.

Literature note 9.3. The NIST DLMF [7] and the foundational paper of Corless et al. [5] tabulate many more branch-point coefficients. The exponential coordinate (57) is particularly convenient in rigorous numerical work, because $t^2/2 = -1 - \log(-x)$ is computed accurately even when x is represented in logarithmic form.

10 Logarithmic asymptotic expansions

The two unbounded real limits are

$$W_0(x) \longrightarrow +\infty \quad (x \to +\infty), \qquad W_{-1}(x) \longrightarrow -\infty \quad (x \to 0^-).$$

They are governed by the same algebra. The first approximation follows by taking logarithms:

$$W + \log|W| = \log|x|.$$

If $|W|$ is large, then W is approximately $\log|x|$, and a second approximation subtracts the logarithm of that first approximation.

10.1 Unsigned Stirling numbers

The unsigned Stirling number of the first kind $\left[{n \atop j}\right]$ is defined by

$$t(t+1)\cdots(t+n-1) = \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix} t^j. \tag{60}$$

It counts permutations of n objects having j cycles, but only its algebraic role is needed here. The recurrence

$$\begin{bmatrix} n \\ j \end{bmatrix} = \begin{bmatrix} n-1 \\ j-1 \end{bmatrix} + (n-1) \begin{bmatrix} n-1 \\ j \end{bmatrix} \tag{61}$$

follows by multiplying the product for $n-1$ by $t+n-1$.

Lemma 10.1 (Exponential generating function)**.** *For $r \geq 0$,*

$$\frac{[-\log(1-z)]^r}{r!} = \sum_{n=r}^{\infty} \begin{bmatrix} n \\ r \end{bmatrix} \frac{z^n}{n!}, \qquad |z| < 1. \tag{62}$$

Consequently,

$$\frac{[-\log(1-z)]^r}{r!(1-z)} = \sum_{n=r}^{\infty} \begin{bmatrix} n+1 \\ r+1 \end{bmatrix} \frac{z^n}{n!}. \tag{63}$$

Proof. Let the right side of (62) be $F_r(z)$. Using (61) and shifting indices shows

$$F_r'(z) = \frac{F_{r-1}(z)}{1-z}, \qquad F_r(0) = 0 \quad (r \geq 1),$$

while $F_0(z) = 1$. The functions $[-\log(1-z)]^r/r!$ satisfy the same differential recurrence and initial conditions, so induction on r proves (62). Differentiate the formula with $r+1$ in place of r to obtain (63). ☐

Define polynomials

$$\mathcal{P}_n(L) = \sum_{m=1}^{n} \frac{(-1)^{n-m}}{m!} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} L^m. \tag{64}$$

The first four are

$$\mathcal{P}_1(L) = L, \tag{65}$$

$$\mathcal{P}_2(L) = \frac{L(-2+L)}{2}, \tag{66}$$

$$\mathcal{P}_3(L) = \frac{L(6-9L+2L^2)}{6}, \tag{67}$$

$$\mathcal{P}_4(L) = \frac{L(-12+36L-22L^2+3L^3)}{12}. \tag{68}$$

Theorem 10.2 (Unified complete real asymptotic expansion)**.** *For either of the following limits,*

branch and limit	L_1	L_2
$W_0(x),\ x \to +\infty$	$\log x$	$\log \log x$
$W_{-1}(x),\ x \to 0^-$	$\log(-x)$	$\log[-\log(-x)]$

one has, for every fixed $N \geq 1$,

$$W(x) = L_1 - L_2 + \sum_{n=1}^{N} \frac{\mathcal{P}_n(L_2)}{L_1^n} + \mathcal{O}\left(\frac{L_2^{N+1}}{|L_1|^{N+1}} \right). \tag{69}$$

Equivalently, the formal complete expansion is

$$W(x) \sim L_1 - L_2 + \sum_{k=0}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^k}{m!} \begin{bmatrix} k+m \\ k+1 \end{bmatrix} \frac{L_2^m}{L_1^{k+m}}. \tag{70}$$

Proof. Set

$$W = L_1 - L_2 + v, \qquad \sigma = \frac{1}{L_1}, \qquad \tau = \frac{L_2}{L_1}.$$

For the principal positive branch, $\log x = W + \log W$. Factoring L_1 from W gives

$$0 = v + \log(1 - \tau + \sigma v).$$

For the negative branch, $\log(-x) = W + \log(-W)$; because $L_1 < 0$, factoring $-L_1$ from $-W$ gives exactly the same equation. Hence

$$\tau = 1 - \mathrm{e}^{-v} + \sigma v. \tag{71}$$

Apply Lagrange inversion to the inverse of the right side:

$$\begin{aligned} v &= \sum_{m=1}^{\infty} \frac{\tau^m}{m} [t^{m-1}] \left(\frac{t}{1 - \mathrm{e}^{-t} + \sigma t} \right)^m \\ &= \sum_{m=1}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k}{m} \binom{m+k-1}{k} \sigma^k \tau^m [t^{m-1}] \left(\frac{t}{1 - \mathrm{e}^{-t}} \right)^{m+k}. \end{aligned} \tag{72}$$

We claim that

$$[t^{m-1}] \left(\frac{t}{1 - \mathrm{e}^{-t}} \right)^{m+k} = \frac{k!}{(m+k-1)!} \begin{bmatrix} m+k \\ k+1 \end{bmatrix}. \tag{73}$$

To prove it, write the coefficient as a formal residue:

$$\operatorname*{Res}_{t=0} \frac{t^k}{(1 - \mathrm{e}^{-t})^{m+k}} \, \mathrm{d}t.$$

Make the substitution $z = 1 - \mathrm{e}^{-t}$, so $t = -\log(1-z)$ and $\mathrm{d}t = \mathrm{d}z/(1-z)$. The residue becomes the coefficient of z^{m+k-1} in

$$\frac{[-\log(1-z)]^k}{1-z}.$$

Formula (63) says that this coefficient is the right side of (73).

Substitution into (72) simplifies the numerical factor:

$$\frac{1}{m} \binom{m+k-1}{k} \frac{k!}{(m+k-1)!} = \frac{1}{m!}.$$

Therefore

$$v = \sum_{m \ge 1} \sum_{k \ge 0} \frac{(-1)^k}{m!} \begin{bmatrix} m+k \\ k+1 \end{bmatrix} \sigma^k \tau^m.$$

Since $\sigma^k \tau^m = L_2^m / L_1^{k+m}$, this is (70); collecting terms with $k + m = n$ gives (64).

Finally, the implicit function in (71) has derivative 1 with respect to v at $(v, \sigma, \tau) = (0,0,0)$. Hence v is analytic in (σ, τ) near the origin. Its Taylor remainder after total degree N is $\mathcal{O}((|\sigma| + |\tau|)^{N+1})$. In both limits, $\sigma \to 0$, $\tau \to 0$, and $|\tau| = L_2/|L_1|$ eventually dominates $|\sigma|$. This proves the remainder estimate in (69). $\square$

10.2 The expansion written separately for each branch

For $x \to +\infty$, set $A = \log x$ and $B = \log A$. Then

$$W_0(x) = A - B + \frac{B}{A} + \frac{B(-2+B)}{2A^2} + \frac{B(6-9B+2B^2)}{6A^3} + \frac{B(-12+36B-22B^2+3B^3)}{12A^4} + \mathcal{O}\left(\frac{B^5}{A^5}\right). \tag{74}$$

For $x \to 0^-$, set

$$\eta = \log \frac{1}{-x}, \qquad B = \log \eta.$$

Since $L_1 = -\eta$,

$$W_{-1}(x) = -\eta - B - \frac{B}{\eta} + \frac{B(B-2)}{2\eta^2} - \frac{B(2B^2-9B+6)}{6\eta^3} + \frac{B(3B^3-22B^2+36B-12)}{12\eta^4} + \mathcal{O}\left(\frac{B^5}{\eta^5}\right). \tag{75}$$

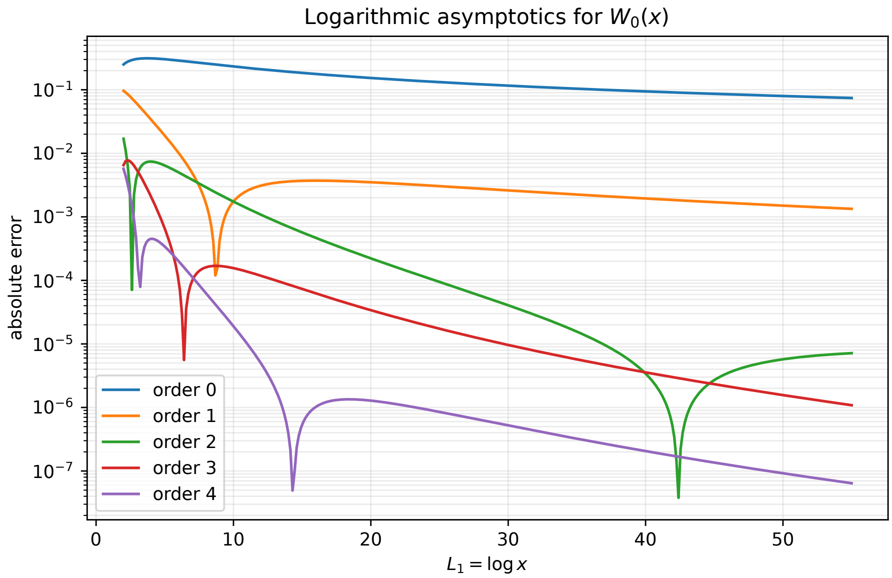

Figure 4: Truncation errors in the logarithmic expansion for W_0. The horizontal coordinate is $A = \log x$, so the right edge represents astronomically large x.

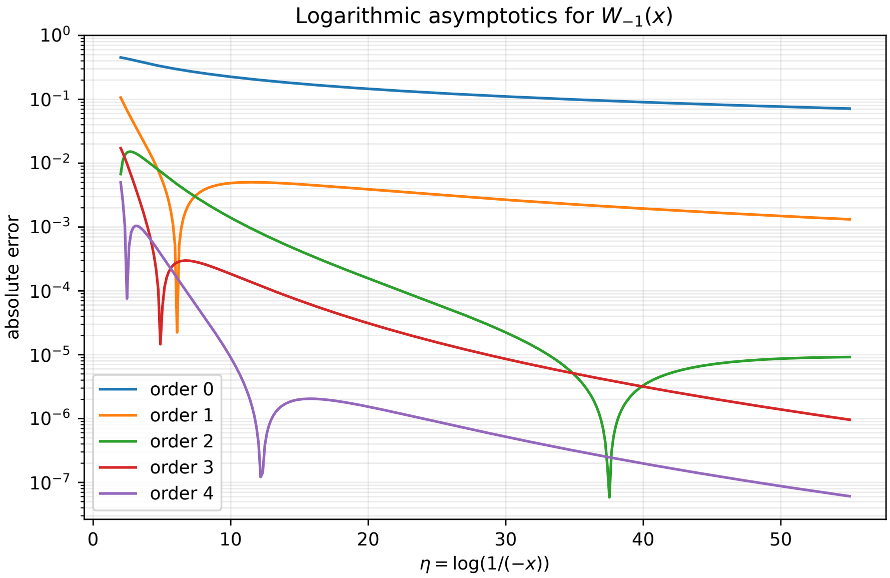

Figure 5: The same coefficient polynomials approximate $W_{-1}(x)$ as $x \to 0^-$, after the sign of L_1 is changed.

Remark 10.3 (Asymptotic does not mean "keep every term"). For a fixed moderate argument, adding terms can eventually stop improving an asymptotic approximation. One should compare successive terms and stop near the smallest one. For genuinely large $|L_1|$, however, each fixed-order truncation in Theorem 10.2 has the proved error order shown there.

11 Rigorous elementary bounds

Bounds serve three purposes: they explain the shape of the graph, provide quick approximations, and furnish guaranteed starting values for iteration.

Lemma 11.1 (Two logarithm inequalities). *For $t > -1$,*

$$\frac{t}{1+t} \le \log(1+t) \le t, \tag{76}$$

with equality only at $t = 0$.

Proof. For the upper bound, let $g(t) = t - \log(1+t)$. Then $g'(t) = t/(1+t)$, so g decreases to $g(0) = 0$ on $(-1, 0)$ and increases from 0 on $(0, \infty)$. Thus $g \ge 0$.

For the lower bound, let $h(t) = \log(1+t) - t/(1+t)$. Then $h'(t) = t/(1+t)^2$. The same monotonicity argument gives $h \ge 0$. □

11.1 Bounds for the principal branch on $x \ge 0$

Theorem 11.2 (Global elementary principal-branch bounds). *For every $x \ge 0$,*

$$\boxed{\frac{x}{1+x} \le W_0(x) \le \log(1+x) \le x.} \tag{77}$$

All inequalities are strict when $x > 0$, except where an accidental equality is forced at $x = 0$.

Proof. Let $w = W_0(x) \geq 0$, so $x = w\mathrm{e}^w$. The elementary inequality $1 - \mathrm{e}^{-w} \leq w$ is equivalent both to

$$\frac{x}{1+x} \leq w$$

and to

$$\mathrm{e}^w \leq 1 + x,$$

the latter being $w \leq \log(1+x)$. Finally $\log(1+x) \leq x$ is lemma 11.1. □

The first rational lower bound is also the $[1/1]$ Pade approximant of W_0 at the origin. Its role is therefore both analytic and numerical.

Theorem 11.3 (Logarithmic bounds for large positive arguments). *Let $x > \mathrm{e}$, $L = \log x > 1$. Then*

$$L - \log L < W_0(x) < L - \log(L - \log L) < L. \tag{78}$$

At $x = \mathrm{e}$, the first lower bound is an equality: $W_0(\mathrm{e}) = 1$.

Proof. The function $f(w) = w\mathrm{e}^w$ is increasing for $w > 0$. Since

$$L\mathrm{e}^L = Lx > x,$$

we have $W_0(x) < L$. Put $a = L - \log L > 0$. Then

$$a\mathrm{e}^a = (L - \log L)\frac{x}{L} = x\left(1 - \frac{\log L}{L}\right) < x,$$

so $a < W_0(x)$. The equation $W_0 = L - \log W_0$ and the lower bound $W_0 > a$ imply

$$W_0 = L - \log W_0 < L - \log a.$$

Finally, the function $h(L) = L - \log L$ is strictly increasing for $L > 1$, and $h(1) = 1$. Thus $a = h(L) > 1$, so $\log a > 0$ and $L - \log a < L$. □

Repeated use of the decreasing map $g(w) = L - \log w$ produces alternating upper and lower bounds. For example, starting with

$$\ell_0 = L - \log L, \qquad u_0 = L,$$

the implication

$$\ell \leq W_0 \leq u \quad \Longrightarrow \quad L - \log u \leq W_0 \leq L - \log \ell$$

follows immediately from the fact that g is decreasing.

11.2 Bounds on the negative part of the principal branch

Theorem 11.4 (Two square-root bounds and a linear bound). *For $-1/\mathrm{e} < x < 0$,*

$$-1 + \sqrt{1 + \mathrm{e}x} < W_0(x) < -1 + \sqrt{2(1 + \mathrm{e}x)}, \tag{79}$$

and also

$$W_0(x) < x. \tag{80}$$

The first lower bound becomes exact at both endpoints in the limiting sense.

Proof. Write $w = -1 + s$ with $0 < s < 1$. Then

$$1 + \mathrm{e}x = 1 + (s-1)\mathrm{e}^s.$$

The lower inequality is equivalent to

$$s^2 > 1 + (s-1)\mathrm{e}^s.$$

Since $s - 1 < 0$, this in turn is equivalent to $1 + s < \mathrm{e}^s$, which is strict for $s > 0$.

For the upper square-root bound, define

$$H(s) = 2[1 - (1-s)\mathrm{e}^s] - s^2.$$

Then $H(0) = 0$ and

$$H'(s) = 2s(\mathrm{e}^s - 1) > 0$$

for $s > 0$. Thus $s^2 < 2(1 + \mathrm{e}x)$, proving the upper bound.

Finally $x = w\mathrm{e}^w$. Since $w < 0$ and $0 < \mathrm{e}^w < 1$, multiplication of the negative number w by e^w moves it toward zero, so $w < x$. □

Near $x = 0$, the bound $W_0(x) < x$ is much sharper than the upper square-root bound; near the branch point, the square-root bounds reveal the correct scale.

11.3 Bounds for the nonprincipal branch near the branch point

The following result, due to Chatzigeorgiou [3], is particularly useful because its proof needs only elementary inequalities.

Theorem 11.5 (Two-sided W_{-1} bound)**.** *For $u > 0$,*

$$\boxed{-1 - \sqrt{2u} - u < W_{-1}(-\mathrm{e}^{-1-u}) < -1 - \sqrt{2u} - \frac{2}{3}u.} \tag{81}$$

Proof. Write

$$W_{-1}(-\mathrm{e}^{-1-u}) = -1 - v, \qquad v > 0.$$

The defining equation gives

$$(1+v)\mathrm{e}^{-v} = \mathrm{e}^{-u},$$

hence

$$u = v - \log(1+v). \tag{82}$$

The desired inequalities are equivalent to

$$\sqrt{2u} + \frac{2}{3}u < v < \sqrt{2u} + u. \tag{83}$$

For the upper bound, $v - u = \log(1+v)$. Put $t = \log(1+v) > 0$, so $v = \mathrm{e}^t - 1$ and $u = \mathrm{e}^t - 1 - t$. The strict exponential inequality $\mathrm{e}^t > 1 + t + t^2/2$ gives

$$t < \sqrt{2u},$$

and hence $v = u + t < u + \sqrt{2u}$.

For the lower bound, note first that $v - \frac{2}{3}u = (v + 2t)/3 > 0$. It is therefore enough to square. The required inequality is equivalent to

$$(v + 2t)^2 > 18u.$$

Substituting $v = \mathrm{e}^t - 1$ and $u = \mathrm{e}^t - 1 - t$, define

$$F(t) = (\mathrm{e}^t - 1 + 2t)^2 - 18(\mathrm{e}^t - 1 - t).$$

A direct differentiation gives

$$F(0) = F'(0) = F''(0) = F'''(0) = 0,$$

and

$$F^{(4)}(t) = 4\mathrm{e}^t(t + 4\mathrm{e}^t - 1) > 0 \qquad (t > 0).$$

Successive integrations from 0 show $F'''(t) > 0$, then $F''(t) > 0$, then $F'(t) > 0$, and finally $F(t) > 0$. This proves the lower inequality in (83), and therefore (81). ☐

Corollary 11.6 (A one-line branch-point approximation)**.** *With $u = -1 - \log(-x) > 0$, define*

$$A_{-1}(x) = -1 - \sqrt{2u} - \frac{5}{6}u. \tag{84}$$

Then

$$|W_{-1}(x) - A_{-1}(x)| < \frac{u}{6}. \tag{85}$$

Proof. The approximation is the midpoint of the two bounds in (81); their distance is $u/3$. ☐

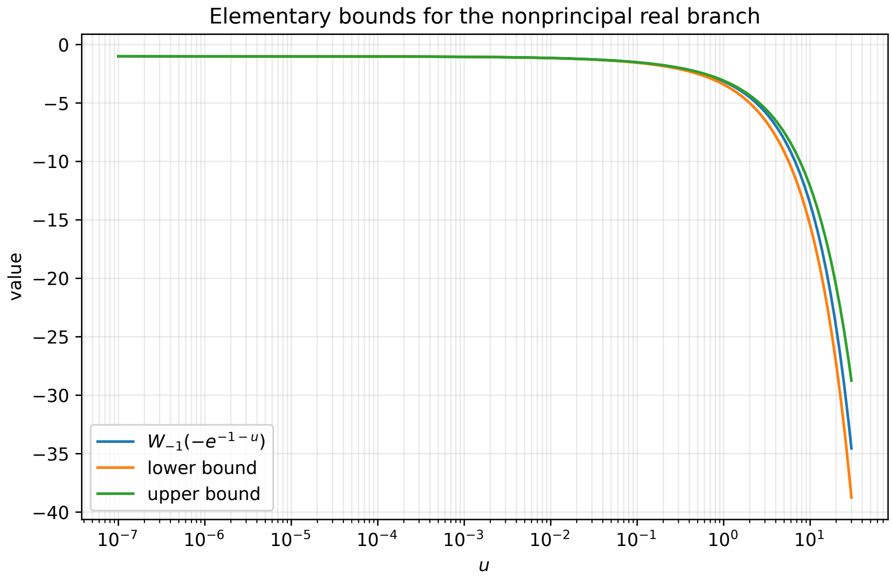

Figure 6: The two elementary bounds of Theorem 11.5. They are especially effective near $u = 0$, which is the branch point.

A simpler square-root consequence, useful when only x is available, is

$$W_{-1}(x) < -1 - \sqrt{2(1 + \mathrm{e}x)}.$$

Indeed, with $w = -1 - v$,

$$v^2 - 2[1 - (1 + v)\mathrm{e}^{-v}]$$

has derivative $2v(1 - \mathrm{e}^{-v}) > 0$ and vanishes at $v = 0$.

11.4 Bounds for W_{-1} near zero

The preceding u-bounds capture the branch point. A different pair captures the logarithmic singularity.

Theorem 11.7 (Near-zero logarithmic bounds)**.** *Let $-1/\mathrm{e} < x < 0$, and put*

$$\eta = \log \frac{1}{-x} > 1.$$

Then

$$-\eta - \frac{\eta \log \eta}{\eta - 1} \le W_{-1}(x) < -\eta - \log \eta. \tag{86}$$

Proof. Let $y = -W_{-1}(x) > 1$. Then

$$y - \log y = \eta, \qquad \text{or} \qquad y = \eta + \log y.$$

Since $y > \eta$, one has

$$y > \eta + \log \eta,$$

which gives the upper bound for $W_{-1} = -y$.

For the other direction, set

$$B = \eta + \frac{\eta \log \eta}{\eta - 1} = \eta + r, \qquad r = \frac{\eta \log \eta}{\eta - 1}.$$

Because $\log(1 + s) \le s$,

$$\log B = \log \eta + \log \left(1 + \frac{r}{\eta}\right) \le \log \eta + \frac{r}{\eta} = r.$$

Thus $B - \log B \ge \eta$. The function $z - \log z$ is strictly increasing for $z > 1$, and $y - \log y = \eta$, so $y \le B$. Multiplying by -1 proves the lower bound in (86). ☐

The interval width in (86) is

$$\frac{\log \eta}{\eta - 1},$$

which tends to zero. Thus the two leading terms $-\eta - \log \eta$ are not merely asymptotically plausible; they are enclosed by an elementary shrinking bracket.

12 Rational and continued-fraction approximations

12.1 Pade approximants at the origin

A $[r/s]$ Pade approximant is a rational function $P_r(x)/Q_s(x)$ whose Taylor series agrees with the target through as many terms as possible, with $Q_s(0) = 1$. Solving the resulting linear coefficient equations for W_0 gives

$$R_{1,1}(x) = \frac{x}{1 + x}, \tag{87}$$

$$R_{2,2}(x) = \frac{2x(4x + 3)}{5x^2 + 14x + 6}, \tag{88}$$

$$R_{3,3}(x) = \frac{3x(451x^2 + 912x + 340)}{665x^3 + 3579x^2 + 3756x + 1020}. \tag{89}$$

Proposition 12.1 (Orders of contact)**.** *At the origin,*

$$W_0(x) - R_{1,1}(x) = \frac{1}{2}x^3 + \mathcal{O}(x^4), \tag{90}$$

$$W_0(x) - R_{2,2}(x) = \frac{17}{72}x^5 + \mathcal{O}(x^6), \tag{91}$$

$$W_0(x) - R_{3,3}(x) = \frac{13489}{122400}x^7 + \mathcal{O}(x^8). \tag{92}$$

None of these three rational functions has a pole on the real domain $[-1/\mathrm{e}, \infty)$ *of* W_0*.*

Proof. Expand each denominator as a geometric power series at zero, multiply by its numerator, and compare with (41). This gives the displayed first unmatched terms. For $R_{1,1}$, the only pole is -1. The roots of the quadratic denominator in $R_{2,2}$ are

$$\frac{-7 \pm \sqrt{19}}{5},$$

and both are less than $-1/\mathrm{e}$.

For the cubic denominator, write

$$D(x) = 665x^3 + 3579x^2 + 3756x + 1020.$$

The exponential series gives $\mathrm{e} > 8/3$, hence $-1/\mathrm{e} > -3/8$. On $[-3/8, \infty)$, the quadratic

$$\frac{D'(x)}{3} = 665x^2 + 2386x + 1252$$

is increasing, because its derivative $1330x + 2386$ is positive there; its value at $-3/8$ is $28849/64 > 0$. Thus D is increasing on that interval, and

$$D(-3/8) = \frac{40821}{512} > 0.$$

It follows that $D(x) > 0$ for every $x \geq -1/\mathrm{e}$. Hence the cubic approximant has no pole on the principal real domain. □

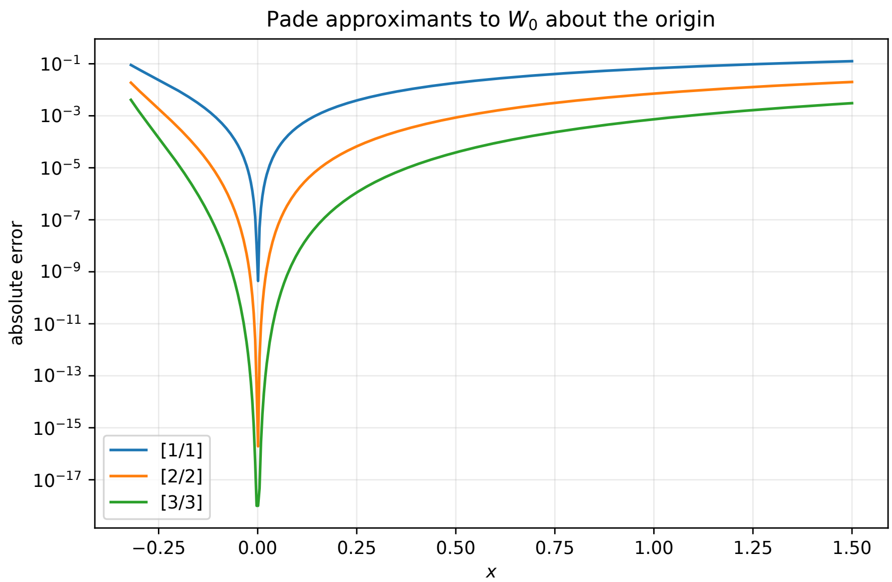

Figure 7: Absolute errors of three diagonal Pade approximants. Their main advantage over a Taylor polynomial is sensible behavior beyond the disk $|x| < 1/\mathrm{e}$.

These approximants belong naturally to W_0. The branch W_{-1} has no Taylor series at $x = 0$; for it, one should first use the branch-point variable p, the exponential variable t, or the logarithmic variable η, and then approximate the transformed function.

12.2 Euler's continued fraction

For a generalized continued fraction, write

$$b_0 + \mathop{\mathbf{K}}_{n=1}^{N} \frac{\alpha_n}{\beta_n} = b_0 + \cfrac{\alpha_1}{\beta_1 + \cfrac{\alpha_2}{\beta_2 + \dots + \cfrac{\alpha_N}{\beta_N}}}.$$

Lemma 12.2 (Euler connection)**.** *Let $c_0, c_1, \dots$ be nonzero where ratios are used, and put $r_n = c_n/c_{n-1}$. The N-th convergent of*

$$c_0 + \frac{c_1}{1 + \mathbf{K}_{n=2}^{N} \frac{-r_n}{1+r_n}} \tag{93}$$

is exactly the partial sum $\sum_{n=0}^{N} c_n$.

Proof. The numerator and denominator continuants P_n, Q_n of a generalized continued fraction obey

$$P_n = \beta_n P_{n-1} + \alpha_n P_{n-2}, \qquad Q_n = \beta_n Q_{n-1} + \alpha_n Q_{n-2}.$$

This recurrence follows by clearing the last denominator in the nested fraction. Here

$$\alpha_1 = c_1, \quad \beta_1 = 1, \qquad \alpha_n = -r_n, \quad \beta_n = 1 + r_n \quad (n \geq 2),$$

with $P_0 = c_0, Q_0 = 1$. The first convergent is $P_1/Q_1 = c_0 + c_1$. If $P_{n-1} = \sum_{j=0}^{n-1} c_j$, $P_{n-2} = \sum_{j=0}^{n-2} c_j$, and $Q_{n-1} = Q_{n-2} = 1$, then

$$P_n = (1 + r_n) \sum_{j=0}^{n-1} c_j - r_n \sum_{j=0}^{n-2} c_j = \sum_{j=0}^{n-1} c_j + r_n c_{n-1} = \sum_{j=0}^{n} c_j,$$

while $Q_n = (1 + r_n) - r_n = 1$. Induction proves the claim. □

For the Lambert series, let

$$c_n = \frac{(-n)^{n-1}}{n!} x^n \qquad (n \geq 1).$$

A calculation gives

$$\frac{c_n}{c_{n-1}} = -\left(1 - \frac{1}{n}\right)^{2-n} x =: a_n x.$$

Theorem 12.3 (Euler continued fraction for W_0)**.** *For $|x| < 1/\mathrm{e}$,*

$$\boxed{W_0(x) = \frac{x}{1 + \mathbf{K}_{n=2}^{\infty} \frac{-a_n x}{1+a_n x}}, \qquad a_n = -\left(1 - \frac{1}{n}\right)^{2-n}.} \tag{94}$$

The convergents are exactly the partial sums of the Maclaurin series.

Proof. Apply lemma 12.2 with $c_0 = 0$ and the displayed ratios. Each convergent equals a Maclaurin partial sum, and those partial sums converge to $W_0(x)$ for $|x| < 1/\mathrm{e}$. □

Literature note 12.4. The paper of Corcino, Corcino, and Mező in the supplied archive develops more sophisticated regular and associated continued fractions, including coefficient formulas through Hankel determinants and numerical studies of their convergence. Formula (94) is the elementary Euler connection from which that discussion begins.

13 Contour and real integral representations

Integral representations are useful for proofs of monotonicity, for quadrature, and for connecting W with complex analysis. We begin with a formula that applies to every branch.

13.1 A local contour representation for all branches

Theorem 13.1 (Inverse-function residue formula)**.** *Let z be a regular value of $f(\zeta) = \zeta e^{\zeta}$, and let Γ_k be a small positively oriented contour enclosing the one simple root $\zeta = W_k(z)$ and no other root of $f(\zeta) = z$. Then, for every integer $m \geq 0$,*

$$W_k(z)^m = \frac{1}{2\pi i} \int_{\Gamma_k} \frac{\zeta^m e^{\zeta}(1+\zeta)}{\zeta e^{\zeta} - z} \, d\zeta. \tag{95}$$

For $m = 1$, this is an integral representation of $W_k(z)$ itself.

Proof. The denominator has a simple zero at $\zeta = W_k(z)$, because $f'(\zeta) = e^{\zeta}(1+\zeta) \neq 0$ at a regular value. The residue of the integrand there is

$$\left. \frac{\zeta^m f'(\zeta)}{f'(\zeta)} \right|_{\zeta = W_k(z)} = W_k(z)^m.$$

The residue theorem proves the formula. □

This representation includes W_{-1} whenever a contour isolates that preimage. What follows is more special: a positive real integral for the principal branch, in the family studied by Kalugin, Jeffrey, and Corless [11].

13.2 The principal-branch cut parametrization

For $0 < v < \pi$, define

$$a(v) = v e^{-v \cot v} \csc v. \tag{96}$$

Lemma 13.2 (Properties of $a(v)$)**.** *The function a is strictly increasing and maps $(0, \pi)$ onto $(1/e, \infty)$. Moreover,*

$$\frac{v a'(v)}{a(v)} = 1 - 2v \cot v + v^2 \csc^2 v = (1 - v \cot v)^2 + v^2 > 0. \tag{97}$$

Proof. Taking logarithms of (96) and differentiating gives

$$\frac{a'}{a} = \frac{1}{v} - 2 \cot v + v \csc^2 v,$$

which is (97). Its final expression is positive.

As $v \to 0$, the expansions $\sin v \sim v$ and $v \cot v \to 1$ give $a(v) \to 1/e$. As $v \to \pi^-$, one has $\csc v \to +\infty$ and $-v \cot v \to +\infty$, so $a(v) \to \infty$. Strict monotonicity now gives the claimed range. □

Why does this function appear? On the upper edge of the principal branch cut, write

$$W_0(-a + i0) = u + iv, \qquad 0 < v < \pi.$$

The condition that $(u + iv)e^{u+iv}$ be negative real forces

$$u = -v \cot v,$$

and its real value is

$$-v e^{-v \cot v} \csc v = -a(v).$$

Thus v parametrizes the entire cut from $-1/e$ to $-\infty$.

Theorem 13.3 (Stieltjes and logarithmic integrals). *For $z \in \mathbb{C} \setminus (-\infty, -1/\mathrm{e}]$,*

$$W_0'(z) = \frac{1}{\pi} \int_0^\pi \frac{\mathrm{d}v}{z + a(v)}, \tag{98}$$

$$W_0(z) = \frac{1}{\pi} \int_0^\pi \operatorname{Log}\left(1 + \frac{z}{a(v)}\right) \mathrm{d}v, \tag{99}$$

$$W_0(z) = \frac{z}{\pi} \int_0^\pi \frac{(1 - v\cot v)^2 + v^2}{z + a(v)} \,\mathrm{d}v. \tag{100}$$

For real $z > -1/\mathrm{e}$, all three integrals are real and the ordinary real logarithm may be used in (99).

Proof. We first prove (98). Let $F(z) = W_0'(z)$. It is analytic off the cut, behaves as $\mathcal{O}(1/z)$ at infinity, and has an integrable inverse-square-root singularity at $-1/\mathrm{e}$. On the upper edge of the cut,

$$W_0(-a(v) + \mathrm{i}0) = -v\cot v + \mathrm{i}v.$$

Differentiating with respect to v, while $z = -a(v)$, gives

$$F(-a(v) + \mathrm{i}0) = -\frac{(-v\cot v)' + \mathrm{i}}{a'(v)}.$$

The lower boundary value is its complex conjugate. Hence the jump is

$$F_+ - F_- = -\frac{2\mathrm{i}}{a'(v)}. \tag{101}$$

Apply Cauchy's formula on a large keyhole contour surrounding the cut. The outer circle contributes zero in the limit because $F(z) = \mathcal{O}(1/z)$; a small circle around the branch point contributes zero because its length is order r and F is order $r^{-1/2}$. The two sides of the cut therefore give

$$F(z) = \frac{1}{2\pi\mathrm{i}} \int_{-\infty}^{-1/\mathrm{e}} \frac{F_+(t) - F_-(t)}{t - z} \,\mathrm{d}t.$$

Set $t = -a(v)$. As t runs from $-\infty$ to $-1/\mathrm{e}$, v runs from π to 0. Using (101), the integral becomes

$$F(z) = \frac{1}{\pi} \int_0^\pi \frac{\mathrm{d}v}{z + a(v)},$$

as claimed.

Integrate this identity from 0 to z along any path in the slit plane. Since $W_0(0) = 0$, and since differentiation under the integral is justified on compact subsets away from the cut,

$$W_0(z) = \frac{1}{\pi} \int_0^\pi \left(\int_0^z \frac{\mathrm{d}s}{s + a(v)} \right) \mathrm{d}v = \frac{1}{\pi} \int_0^\pi \operatorname{Log}\left(1 + \frac{z}{a(v)}\right) \mathrm{d}v.$$

This proves (99).

Finally let $h(v) = \operatorname{Log}(1 + z/a(v))$. The boundary term $[vh(v)]_0^\pi$ vanishes: the factor v handles $v = 0$, and $a(v) \to \infty$ handles $v = \pi$. Integration by parts gives

$$\int_0^\pi h(v)\,\mathrm{d}v = z \int_0^\pi \frac{v a'(v)}{a(v)[z + a(v)]} \,\mathrm{d}v.$$

Insert (97) to obtain (100). □

Corollary 13.4 (Alternating signs of all principal-branch derivatives). *For every integer $n \geq 0$ and every real $x > -1/\mathrm{e}$,*

$$(-1)^n W_0^{(n+1)}(x) = \frac{n!}{\pi} \int_0^\pi \frac{\mathrm{d}v}{[x + a(v)]^{n+1}} > 0. \tag{102}$$

Thus W_0' is completely monotone: $W_0' > 0$, $W_0'' < 0$, $W_0''' > 0$, and so on.

Proof. Differentiate (98) n times under the integral. For $x > -1/\mathrm{e}$, lemma 13.2 gives $x + a(v) > 0$, so the integrand on the right of (102) is positive. □

This all-orders sign property is special to the principal branch. It cannot extend to W_{-1}, whose second derivative changes sign at $-2/\mathrm{e}^2$ by Theorem 5.4.

Corollary 13.5 (An integral for rooted-tree numbers)**.** *For every integer $m \geq 1$,*

$$\frac{1}{\pi}\int_0^{\pi} a(v)^{-m}\,\mathrm{d}v = \frac{m^{m-1}}{(m-1)!}. \tag{103}$$

Proof. Set $x = 0$ and $n = m - 1$ in (102). The left side is

$$(-1)^{m-1} W_0^{(m)}(0) = m^{m-1}$$

by (43). Divide by $(m-1)!$. □

Expanding the logarithm in (99) and using (103) gives a second proof of the Maclaurin series:

$$\frac{1}{\pi}\int_0^{\pi} \sum_{m\geq 1} \frac{(-1)^{m-1}}{m}\frac{z^m}{a(v)^m}\,\mathrm{d}v = \sum_{m\geq 1}\frac{(-m)^{m-1}}{m!}z^m.$$

This identity ties together three representations that at first seem unrelated: rooted labelled trees, a branch-cut parametrization, and the inverse of $w\mathrm{e}^w$.

14 Numerical evaluation

A computer does not evaluate a special function by magic. It combines:

1. a representation adapted to the part of the domain,
2. a starting approximation,
3. a rapidly convergent correction,
4. and a reliable stopping or error test.

The branch point, the origin, and large arguments require different representations.

14.1 Residuals give certified a posteriori errors

Theorem 14.1 (Residual error bound)**.** *Let $r = W_k(x)$, let $\widehat{w}$ be an approximation on the same real inverse interval, and let I be the closed interval joining r and $\widehat{w}$. If I does not contain -1, put*

$$m_I = \min_{t\in I}\left|\mathrm{e}^t(1+t)\right|.$$

Then

$$|\widehat{w} - r| \leq \frac{\left|\widehat{w}\mathrm{e}^{\widehat{w}} - x\right|}{m_I}. \tag{104}$$

For $x \neq 0$, an overflow-resistant alternative uses

$$G_x(w) = w + \log|w| - \log|x|.$$

If, in addition, I avoids zero and

$$\mu_I = \min_{t\in I}|1 + 1/t| > 0,$$

then

$$|\widehat{w} - r| \leq \frac{|G_x(\widehat{w})|}{\mu_I}. \tag{105}$$

Proof. Apply the mean value theorem to $f(t) = t\mathrm{e}^t - x$:

$$f(\widehat{w}) - f(r) = f'(\xi)(\widehat{w} - r)$$

for some $\xi \in I$. Since $f(r) = 0$ and $|f'(\xi)| \geq m_I$, (104) follows. The proof of (105) is identical, using $G_x(r) = 0$ and $G_x'(t) = 1 + 1/t$. □

At the branch point m_I and μ_I can be small. This is not a defect of the estimate: Equation (25) shows that the inverse problem itself is ill-conditioned there. The signed variable p is the correct coordinate in that regime.

14.2 Newton and Halley iterations

For

$$f_x(w) = w\mathrm{e}^w - x,$$

Newton's method is

$$w^+ = w - \frac{w\mathrm{e}^w - x}{\mathrm{e}^w(1+w)} = \frac{w^2 + x\mathrm{e}^{-w}}{1+w}. \tag{106}$$

Halley's method is

$$w^+ = w - \frac{2f_x(w)f_x'(w)}{2f_x'(w)^2 - f_x(w)f_x''(w)}, \qquad f_x''(w) = \mathrm{e}^w(w+2). \tag{107}$$

Theorem 14.2 (Local orders of convergence)**.** *Let r be a simple root of a three-times continuously differentiable function f, so $f(r) = 0$ and $f'(r) \neq 0$. If $w = r + e$ is sufficiently close to r, Newton's method has*

$$e^+ = \frac{f''(r)}{2f'(r)}e^2 + \mathcal{O}(e^3),$$

while Halley's method has $e^+ = \mathcal{O}(e^3)$. Applied to the Lambert equation away from $r = -1$, Newton is locally quadratic and Halley locally cubic.

Proof. Write $a = f'(r)$, $b = f''(r)$, $c = f'''(r)$. Taylor expansion gives

$$f(r+e) = ae + \frac{b}{2}e^2 + \frac{c}{6}e^3 + \mathcal{O}(e^4),$$

$$f'(r+e) = a + be + \frac{c}{2}e^2 + \mathcal{O}(e^3).$$

Division yields

$$\frac{f(r+e)}{f'(r+e)} = e - \frac{b}{2a}e^2 + \mathcal{O}(e^3),$$

which proves the Newton formula.

For Halley, substitute the same expansions into

$$\frac{2ff'}{2(f')^2 - ff''}.$$

Both numerator and denominator have the same relative linear term, so their quotient is $e + \mathcal{O}(e^3)$; the e^2 term cancels. Subtracting this correction from e leaves $\mathcal{O}(e^3)$. □

The root at the common branch point is not simple: $f'(-1) = 0$. One should return -1 exactly when $x = -1/\mathrm{e}$, or use the square-root expansion for nearby inputs.

14.3 Logarithmic Newton iteration

For nonzero x, solve instead

$$G_x(w) = w + \log|w| - \log|x| = 0.$$

Newton's formula simplifies to

$$\boxed{N_x(w) = \frac{w}{1+w}\left(1 + \log\left|\frac{x}{w}\right|\right).} \tag{108}$$

This form is central in guaranteed high-precision algorithms because it avoids forming e^w.

We first prove a global one-step ordering theorem.

Theorem 14.3 (Branch-safe monotone Newton step). *Let $r = W_k(x)$, and suppose w has the same sign as x.*

(i) *If $0 < w < r$, then $w < N_x(w) < r$.*
(ii) *If $-1 < r < w < 0$, then $r < N_x(w) < w$.*
(iii) *If $w < r < -1$, then $w < N_x(w) < r$.*

Proof. Put $t = r/w > 0$. Since $\log|x| = r + \log|r|$, direct algebra gives the exact identities

$$N_x(w) - w = \frac{w}{1+w}\left[w(t-1) + \log t\right], \tag{109}$$

$$r - N_x(w) = \frac{w}{1+w}\left[t - 1 - \log t\right]. \tag{110}$$

For every $t > 0$, $t \neq 1$,

$$\frac{t-1}{t} < \log t < t - 1, \tag{111}$$

which is lemma 11.1 after a change of variable.

In case (i), $t > 1$, $w/(1+w) > 0$, and both brackets in (109) and (110) are positive. Hence $w < N_x(w) < r$.

In case (ii), $t > 1$ and $w/(1+w) < 0$. Since $r = wt > -1$, $w > -1/t$, and therefore

$$w(t-1) + \log t > -\frac{t-1}{t} + \log t > 0.$$

Thus (109) gives $N_x(w) < w$, while (110) gives $r < N_x(w)$.

In case (iii), $0 < t < 1$ and $w/(1+w) > 0$. Since $r = wt < -1$, one has $w < -1/t$. Multiplying by the negative number $t - 1$ gives

$$w(t-1) > \frac{1-t}{t}.$$

The left inequality of (111) is $\log t > -(1-t)/t$, so the bracket in (109) is positive. The bracket in (110) is always positive. Hence $w < N_x(w) < r$. □

Theorem 14.4 (Elementary globally convergent starts). *For a real input in the stated branch domain, choose*

$$w_0 = \begin{cases} x/(1+x), & k = 0, \quad 0 < x < \mathrm{e}, \\ \log x - \log\log x, & k = 0, \quad x \geq \mathrm{e}, \\ x, & k = 0, \quad -1/\mathrm{e} < x < 0, \\ -1 - \sqrt{2u} - u, & k = -1, \quad -1/\mathrm{e} < x < 0, \quad u = -1 - \log(-x). \end{cases} \tag{112}$$

Then $w_{n+1} = N_x(w_n)$ is well defined, remains on the selected branch interval, is monotone, and converges to $W_k(x)$.

Proof. For $0 < x < \mathrm{e}$, Theorem 11.2 gives $0 < w_0 < W_0(x)$, so case (i) of Theorem 14.3 applies. For $x > \mathrm{e}$, Theorem 11.3 gives the same ordering; at $x = \mathrm{e}$, the start is the exact value 1.

For $-1/\mathrm{e} < x < 0$ on W_0, (80) gives $-1 < W_0(x) < w_0 = x < 0$, so case (ii) applies. On W_{-1}, Theorem 11.5 gives $w_0 < W_{-1}(x) < -1$, so case (iii) applies.

In each case the sequence is monotone and bounded by the root, so it has a limit ℓ in the same interval. Continuity gives $N_x(\ell) = \ell$. Since $\ell \neq 0, -1$, this fixed-point equation reduces to

$$\log |x/\ell| = \ell,$$

or $x = \ell \mathrm{e}^{\ell}$. Uniqueness on the selected inverse interval gives $\ell = W_k(x)$. ☐

Corollary 14.5 (Quadratic error constant)**.** *Let $r = W_k(x) \neq -1, 0$, and let $e_n = w_n - r$ for the logarithmic Newton iteration. Then*

$$e_{n+1} = -\frac{1}{2r(1+r)} e_n^2 + \mathcal{O}(e_n^3). \tag{113}$$

Proof. Apply the Newton error formula of Theorem 14.2 to $G_x'(w) = 1 + 1/w$ and $G_x''(w) = -1/w^2$. ☐

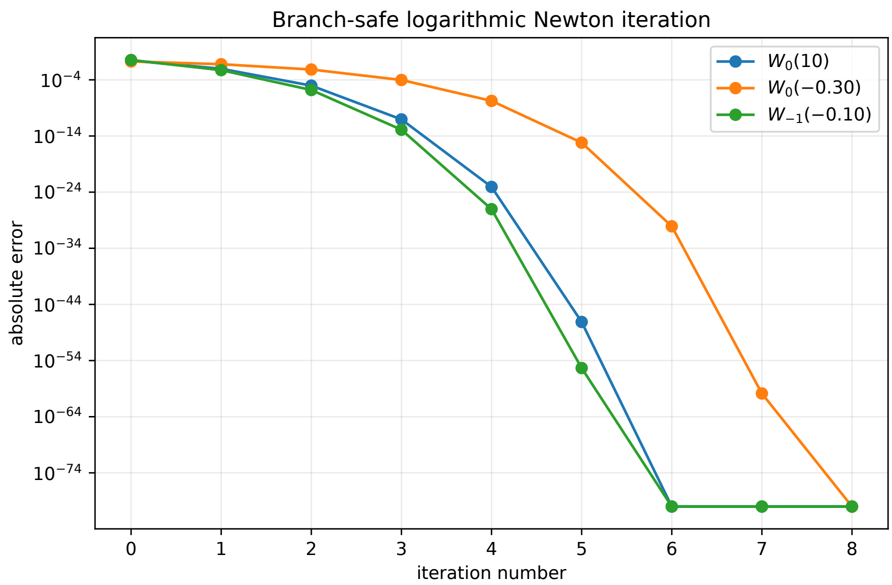

Figure 8: Monotone, eventually quadratic convergence of (108) in three different branch regimes.

14.4 A practical branch-aware algorithm

For ordinary floating-point work, the following strategy is robust.

Algorithm 14.6 (Real Lambert W)**.** *Given a real x and branch $k \in \{0, -1\}$:*

1. *Reject inputs outside the branch domain. Return -1 exactly at $x = -1/\mathrm{e}$, and return 0 for $W_0(0)$.*
2. *If $1 + \mathrm{e}x$ is tiny, evaluate the signed Puiseux series (51).*
3. *If $k = 0$ and $|x|$ is small, use the Maclaurin series or a Pade approximant.*
4. *If $k = 0$ and x is large, use (74) as a start. If $k = -1$ and x is close to zero, use (75).*
5. *Otherwise use a guaranteed start from (112).*
6. *Apply logarithmic Newton or Halley until the logarithmic residual and the change in w are both below the requested tolerance.*

If the input $x < 0$ is so small that it underflows, store $\lambda = \log(-x)$ instead. Formula (108) then becomes

$$N_\lambda(w) = \frac{w}{1+w}\left(1 + \lambda - \log(-w)\right), \tag{114}$$

which never needs x itself.

Literature note 14.7. The 2022 paper by Lajos Lőczi [13] in the supplied archive develops guaranteed-precision versions of logarithmic Newton iteration, including piecewise starting values and explicit error estimates over the full real domains. The paper by Iacono and Boyd [10] constructs global analytic approximations and iteration-based refinements for W_0. Howard's Schröder-series work [9] studies higher-order root-solving expansions. The present article emphasizes a shorter set of starts for which monotone convergence can be proved directly by Theorem 14.3.

14.5 Which approximation should be used?

Regime	Recommended representation	Natural small parameter
W_0, $x \approx 0$	Maclaurin series or Pade approximant	x
Either branch, $x \approx -1/\mathrm{e}$	signed Puiseux series	$p = \pm\sqrt{2(1+\mathrm{e}x)}$
W_0, $x \gg 1$	logarithmic asymptotic series	$(\log\log x)/\log x$
W_{-1}, $x \to 0^-$	logarithmic asymptotic series	$\log\eta/\eta$
Moderate x, high accuracy	logarithmic Newton or Halley	current residual
Certified value	monotone bracket plus residual bound	bracket width

There is no single best formula. A high-quality implementation is a carefully organized collection of mathematically equivalent formulas, each used where it is well conditioned.

15 Solving transcendental equations

The main skill is to isolate an expression of the form

$$Y\mathrm{e}^Y = Z.$$

Branch selection is then decided by the allowable range of the original unknown, not by algebra alone.

15.1 A linear term plus a logarithm

Theorem 15.1. *Let $a, b, c \in \mathbb{R}$, with $ab \neq 0$, and seek $x > 0$ satisfying*

$$ax + b\log x = c.$$

Every real solution is

$$x = \frac{b}{a} W_k\left(\frac{a}{b}\mathrm{e}^{c/b}\right), \tag{115}$$

where a real branch is chosen so that the right side is positive.

Proof. Rearrange and exponentiate:

$$x = \mathrm{e}^{c/b}\mathrm{e}^{-(a/b)x}.$$

Therefore

$$\frac{a}{b}x\exp\left(\frac{a}{b}x\right) = \frac{a}{b}\mathrm{e}^{c/b}.$$

Apply a real branch of W, then multiply by b/a. Every reversible step preserves all positive solutions. □

Two important special cases are

$$x + \log x = c \quad \Longrightarrow \quad x = W_0(\mathrm{e}^c),$$

and

$$x - \log x = c \quad \Longrightarrow \quad x = -W_k(-\mathrm{e}^{-c}). \tag{116}$$

For $c > 1$, (116) has two positive solutions: $-W_0(-\mathrm{e}^{-c}) \in (0, 1)$ and $-W_{-1}(-\mathrm{e}^{-c}) \in (1, \infty)$. At $c = 1$, they coalesce at $x = 1$. This is exactly the two-branch geometry of Equation (5).

15.2 A power times an exponential

Theorem 15.2. *Let $p \neq 0$, $q \neq 0$, and suppose a consistent real p-th root of A has been selected. Solutions of*

$$x^p \mathrm{e}^{qx} = A$$

that are compatible with that root are

$$x = \frac{p}{q} W_k\left(\frac{q}{p} A^{1/p}\right). \tag{117}$$

Proof. Taking the selected p-th root gives $x\mathrm{e}^{(q/p)x} = A^{1/p}$. Multiplication by q/p produces the defining Lambert equation. Conversely, substitution verifies every value in (117). □

When p is even, taking a real root can introduce two signs. Each sign must be treated separately, and each resulting Lambert argument may itself admit two real branches.

Remark 15.3 (Endpoint-safe integer-power inverse package). For $m \in \mathbb{N}_0 \setminus \{0\}$, $A > 0$, and $\beta > 0$, the project sets

$$S_{m,A,\beta}(t) = At^m \mathrm{e}^{-\beta t}, \qquad P_{m,A,\beta} = A\left(\frac{m}{\beta}\mathrm{e}^{-1}\right)^m$$

and defines principal and lower Lambert phases λ_0 and λ_{-1}. Their exact endpoint-inclusive images are

$$\lambda_0\big([0, P_{m,A,\beta}]\big) = \left[0, \frac{m}{\beta}\right], \qquad \lambda_{-1}\big((0, P_{m,A,\beta}]\big) = \left[\frac{m}{\beta}, \infty\right].$$

These are `principalPowerExponentialPhase_image_Icc` and `lowerPowerExponentialPhase_image_Ioc`. The theorems `principalPowerExponentialPhase_invOn` and `lowerPowerExponentialPhase_invOn` further say that each phase and $S_{m,A,\beta}$ are two-sided setwise inverses on the corresponding displayed interval pair. All four declarations are in `PowerExponentialLambertInverse.lean`. Thus this is a branch-explicit, endpoint-safe integer-power specialization of Theorem 15.2, without making an arbitrary-real-power root claim.

The normalized Lambert argument is continuous by `powerExponentialLambertArgument_continuous`; the full phase statements are `principalPowerExponentialPhase_continuousOn_Icc` on $[0, P_{m,A,\beta}]$ and `lowerPowerExponentialPhase_continuousOn_Ioc` on $(0, P_{m,A,\beta}]$. Moreover, if $0 < x \leq P_{m,A,\beta}$ and $\lambda \geq 0$, then the exact equivalence

$$S_{m,A,\beta}(\lambda) = x \quad \Longleftrightarrow \quad \lambda = \lambda_0(x) \text{ or } \lambda = \lambda_{-1}(x)$$

is `powerExponentialSaddle_eq_iff_eq_principal_or_eq_lower`. Strictly inside the value interval the alternatives are distinct, by `principalPowerExponentialPhase_ne_lowerPowerExponentialPhase`. The nonnegativity restriction is essential when even m permits additional negative roots.

One further differentiation is formalized in `PowerExponentialLambertCurvature.lean`. For either $k \in \{0, -1\}$ and every $0 < x < P_{m,A,\beta}$, it gives

$$\lambda_k''(x) = \frac{\lambda_k(x)\left[m - \big(m - \beta\lambda_k(x)\big)^2\right]}{x^2\big(m - \beta\lambda_k(x)\big)^3}. \tag{118}$$

The derivative witnesses are `deriv_principalPowerExponentialPhase_hasDerivAt` and `deriv_lowerPowerExponentialPhase_hasDerivAt`; the corresponding exact formulas are `deriv_deriv_principalPowerExponentialPhase` and `deriv_deriv_lowerPowerExponentialPhase`. This generic module is deliberately formula-only. The sign and zero classifications at $\beta\lambda_0 = m - \sqrt{m}$ and $\beta\lambda_{-1} = m + \sqrt{m}$, and the resulting generic strict-shape theorems, are not yet packaged as Lean declarations.

The current zero-endpoint asymptotic package is exactly `PowerExponentialLambertAsymptotics.lean`. It proves `principalPowerExponentialPhase_isEquivalent_rpow`, namely

$$\lambda_0(x) \sim \left(\frac{x}{A}\right)^{1/m} \qquad (x \downarrow 0),$$

and `tendsto_lowerPowerExponentialPhase_nhdsGT_zero_atTop`, namely $\lambda_{-1}(x) \to +\infty$. For the intrinsic variable

$$\varepsilon(x) = \frac{\beta}{m}\left(\frac{x}{A}\right)^{1/m},$$

`lowerPowerExponentialPhase_sub_intrinsicMain_tendsto_zero` gives only the two-term statement

$$\lambda_{-1}(x) + \frac{m}{\beta}\big(\log\varepsilon(x) - \log|\log\varepsilon(x)|\big) \longrightarrow 0.$$

No cleaned normalization in $L = \log(A/x)$, nor a full generic asymptotic series, is claimed by that module.

At $(m, A, \beta) = (1, 1, \log 2)$, the profile is $t2^{-t}$, its peak is $\mathrm{e}^{-1}/\log 2$, and the principal specialization is `fabiusPrincipalLambertPhase`. The exact two-root equivalence and strict interior distinctness become `fabiusSaddle_eq_iff_eq_principal_or_eq_lower` and `fabiusPrincipalLambertPhase_ne_fabiusLambertPhase`; continuity on the closed principal and positive endpoint-inclusive lower value intervals is `fabiusPrincipalLambertPhase_continuousOn_Icc` and `fabiusLambertPhase_continuousOn_Ioc`. Finally, `fabiusPrincipalLambertPhase_isEquivalent_id` states that the principal Fabius phase is asymptotic to x at zero, while `lowerPowerExponentialPhase_one_one_log_two` in `PowerExponentialLambertFabius.lean` identifies the lower generic phase exactly with the Fabius lower-Lambert phase.

The separate specialization `PowerExponentialLambertFabiusCurvature.lean` packages the complete classical curvature geometry. Put

$$L = \log 2, \qquad P = \frac{\mathrm{e}^{-1}}{L}, \qquad x_* = \frac{2\mathrm{e}^{-2}}{L}.$$

The definition `fabiusLambertInflectionInput` is x_*, `fabiusLambertInflectionInput_mem_Ioo` proves $0 < x_* < P$, and `fabiusLambertPhase_inflectionInput` gives $\lambda_{-1}(x_*) = 2/L$. On $0 < x < P$, `deriv_deriv_fabiusLambertPhase` states

$$\lambda_{-1}''(x) = \frac{L\lambda_{-1}(x)^2\big(2 - L\lambda_{-1}(x)\big)}{x^2\big(1 - L\lambda_{-1}(x)\big)^3}. \tag{119}$$

The declarations `deriv_deriv_fabiusLambertPhase_pos_iff`, `deriv_deriv_fabiusLambertPhase_neg_iff`, and `deriv_deriv_fabiusLambertPhase_eq_zero_iff` say respectively that its sign is positive for $x < x_*$, negative for $x_* < x$, and zero exactly at $x = x_*$; the named zero is `deriv_deriv_fabiusLambertPhase_inflectionInput`. Consequently `strictConvexOn_fabiusLambertPhase_left` gives strict convexity on $(0, x_*]$, while `strictConcaveOn_fabiusLambertPhase_right` gives strict concavity on $[x_*, P]$.

The principal specialization is global to the left of the peak, not merely restricted to positive profile inputs. The bridge `fabiusPrincipalLambertPhase_eq_principalLambertW` is

$$\lambda_0(x) = -\frac{1}{L} W_0(-Lx).$$

The declarations `fabiusPrincipalLambertPhase_continuousOn_Iic`, `fabiusPrincipalLambertPhase_hasDerivAt`, and `deriv_fabiusPrincipalLambertPhase` give continuity on $(-\infty, P]$ and first-order calculus on $(-\infty, P)$. The derivative witness `deriv_fabiusPrincipalLambertPhase_hasDerivAt` and exact formula `deriv_deriv_fabiusPrincipalLambertPhase` give, for $x < P$,

$$\lambda_0''(x) = L \frac{\mathrm{e}^{-2W_0(-Lx)} \bigl(W_0(-Lx) + 2\bigr)}{\bigl(W_0(-Lx) + 1\bigr)^3} > 0. \tag{120}$$

Strict positivity is `deriv_deriv_fabiusPrincipalLambertPhase_pos`, the regular value $\lambda_0''(0) = 2L$ is `deriv_deriv_fabiusPrincipalLambertPhase_zero`, and `strictConvexOn_fabiusPrincipalLambertPhase` proves strict convexity on the whole closed half-line $(-\infty, P]$.

15.3 An exponential equal to an affine function

Theorem 15.4. *Let $\alpha\beta \neq 0$. Every solution of*

$$\mathrm{e}^{\alpha x} = \beta x + \gamma \tag{121}$$

is

$$\boxed{x = -\frac{\gamma}{\beta} - \frac{1}{\alpha} W_k\left(-\frac{\alpha}{\beta}\mathrm{e}^{-\alpha\gamma/\beta}\right).} \tag{122}$$

Real solutions correspond exactly to real branches available at the displayed argument.

Proof. Set $y = \beta x + \gamma$, so $x = (y - \gamma)/\beta$. Equation (121) becomes

$$\mathrm{e}^{\alpha(y-\gamma)/\beta} = y.$$

Multiplying by $\exp(-\alpha y/\beta)$ and then by $-\alpha/\beta$ gives

$$\left(-\frac{\alpha y}{\beta}\right) \exp\left(-\frac{\alpha y}{\beta}\right) = -\frac{\alpha}{\beta}\mathrm{e}^{-\alpha\gamma/\beta}.$$

Apply W_k, solve for y, and then for x. □

Example 15.5. *The equation $\mathrm{e}^x = 3x$ has argument $-1/3 \in (-1/\mathrm{e}, 0)$, so it has two positive solutions:*

$$x = -W_0(-1/3), \qquad x = -W_{-1}(-1/3).$$

The two intersections of an exponential and a line are precisely the two real Lambert branches.

15.4 The equation $x^x = A$

Theorem 15.6 (All positive solutions of $x^x = A$)**.** *Let $A > 0$. Positive solutions of $x^x = A$ are*

$$x = \exp(W_k(\log A)). \tag{123}$$

Their number is:

- *one if $A > 1$;*
- *one if $A = 1$, namely $x = 1$;*
- *two if $\mathrm{e}^{-1/\mathrm{e}} < A < 1$;*
- *one double minimum solution if $A = \mathrm{e}^{-1/\mathrm{e}}$, namely $x = \mathrm{e}^{-1}$;*
- *none if $0 < A < \mathrm{e}^{-1/\mathrm{e}}$.*

Proof. Put $y = \log x$, so $x = \mathrm{e}^y$. Taking logarithms of $x^x = A$ gives

$$y\mathrm{e}^y = \log A.$$

Thus $y = W_k(\log A)$, proving (123). If $\log A > 0$, only W_0 is real. If $-1/\mathrm{e} < \log A < 0$, both W_0 and W_{-1} are real. At $\log A = -1/\mathrm{e}$, they meet. Below that value no real branch exists.

For a direct calculus check, the derivative of $\log(x^x) = x \log x$ is $1 + \log x$, so x^x has its positive minimum at $x = \mathrm{e}^{-1}$, with value $\mathrm{e}^{-1/\mathrm{e}}$. □

15.5 Fixed points of exponentiation

A positive fixed point of

$$y = x^y$$

satisfies

$$y = -\frac{W_k(-\log x)}{\log x} \qquad (x \neq 1). \tag{124}$$

Proof. Taking logarithms gives $\log y = y \log x$. Set $z = -y \log x$. Then

$$z\mathrm{e}^z = (-y \log x)\mathrm{e}^{-y \log x} = (-y \log x)\frac{1}{y} = -\log x.$$

Thus $z = W_k(-\log x)$, which gives (124). □

For $0 < x < 1$, the argument $-\log x$ is positive and there is one positive fixed point. For $1 < x < \mathrm{e}^{1/\mathrm{e}}$, the argument lies in $(-1/\mathrm{e}, 0)$, so there are two positive fixed points. At $x = \mathrm{e}^{1/\mathrm{e}}$, they merge at $y = \mathrm{e}$. This statement concerns solutions of the fixed-point equation; convergence of the iterated power tower is a separate dynamical question.

15.6 A Planck-type maximization problem

Theorem 15.7. *Let $n > 1$. The function*

$$F_n(x) = \frac{x^n}{\mathrm{e}^x - 1}, \qquad x > 0,$$

has a unique critical point, which is its global maximum, at

$$x_n = n + W_0(-n\mathrm{e}^{-n}). \tag{125}$$

The second real branch gives

$$n + W_{-1}(-n\mathrm{e}^{-n}) = 0,$$

the boundary point rather than an interior maximum.

Proof. Logarithmic differentiation gives

$$\frac{F_n'(x)}{F_n(x)} = \frac{n}{x} - \frac{\mathrm{e}^x}{\mathrm{e}^x - 1}.$$

A critical point satisfies

$$n(\mathrm{e}^x - 1) = x\mathrm{e}^x,$$

or

$$(n - x)\mathrm{e}^x = n.$$

Put $y = x - n$. Then

$$y\mathrm{e}^y = -n\mathrm{e}^{-n},$$

so $x = n + W_k(-n\mathrm{e}^{-n})$. Since $-n \le -1$, corollary 2.4 gives $W_{-1}(-n\mathrm{e}^{-n}) = -n$, hence the boundary value $x = 0$. The principal value lies in $(-1, 0)$, so $x_n \in (n-1, n)$.

Finally $F_n(x) \to 0$ both as $x \downarrow 0$ (because $n > 1$) and as $x \to \infty$. The critical-point equation has only the one interior solution just found, so that solution is the global maximum. □

16 Applications

The point of a special function is not merely to rename an equation. A closed form exposes parameter dependence, branch multiplicity, asymptotic regimes, and reliable initial values for computation.

16.1 Michaelis–Menten substrate depletion

Consider

$$\frac{\mathrm{d}S}{\mathrm{d}t} = -\frac{V_{\max} S}{K_M + S}, \qquad S(0) = S_0 > 0. \tag{126}$$

Theorem 16.1. *The solution is*

$$S(t) = K_M W_0\left(\frac{S_0}{K_M} \exp\left[\frac{S_0}{K_M} - \frac{V_{\max} t}{K_M}\right]\right). \tag{127}$$

Proof. Separate variables:

$$\left(1 + \frac{K_M}{S}\right) \mathrm{d}S = -V_{\max}\,\mathrm{d}t.$$

Integration and the initial condition give

$$S + K_M \log S = S_0 + K_M \log S_0 - V_{\max} t.$$

Put $y = S/K_M$ and $y_0 = S_0/K_M$. Constants involving $\log K_M$ cancel, leaving

$$y + \log y = y_0 + \log y_0 - \frac{V_{\max} t}{K_M}.$$

Exponentiation gives

$$y\mathrm{e}^y = y_0 \exp\left(y_0 - \frac{V_{\max} t}{K_M}\right).$$

The right side is positive, so the only real branch is W_0, proving (127). □

The archive paper by Goličnik [8] develops this and related biochemical kinetic examples in detail. Formula (127) makes it possible to differentiate with respect to parameters without repeatedly solving an implicit equation.

16.2 A diode with a series resistor

Let $I_S > 0$ be the saturation current, $R_S > 0$ the series resistance, and $nV_T > 0$ the thermal-voltage scale. The Shockley law and the circuit voltage relation are

$$I = I_S\left(\mathrm{e}^{V_D/(nV_T)} - 1\right), \qquad V = V_D + IR_S.$$

Theorem 16.2. *The current is*

$$I = \frac{nV_T}{R_S} W_0\left(\frac{R_S I_S}{nV_T}\exp\left[\frac{V + R_S I_S}{nV_T}\right]\right) - I_S. \tag{128}$$

Proof. Let $Y = I + I_S$. Substitution of $V_D = V - IR_S$ gives

$$Y = I_S \exp\left(\frac{V - R_S(Y - I_S)}{nV_T}\right).$$

Therefore

$$Y \exp\left(\frac{R_S Y}{nV_T}\right) = I_S \exp\left(\frac{V + R_S I_S}{nV_T}\right).$$

Multiplication by $R_S/(nV_T)$ produces a positive Lambert argument, so W_0 is the unique real branch. Solving for $I = Y - I_S$ gives (128). □

This branch-explicit form underlies the transistor and diode analyses in Banwell's paper [1].

16.3 Characteristic exponents of a delay equation

Consider

$$y'(t) = a\,y(t-\tau), \qquad \tau > 0. \tag{129}$$

Theorem 16.3. *Exponential modes $y(t) = \mathrm{e}^{\lambda t}$ have exponents*

$$\lambda_k = \frac{1}{\tau} W_k(a\tau), \qquad k \in \mathbb{Z}. \tag{130}$$

Proof. Substitution gives

$$\lambda \mathrm{e}^{\lambda t} = a\mathrm{e}^{\lambda(t-\tau)},$$

so $\lambda = a\mathrm{e}^{-\lambda\tau}$. Hence

$$(\lambda\tau)\mathrm{e}^{\lambda\tau} = a\tau,$$

and (130) follows. □

Here the infinitely many complex branches are not optional decoration: they form the characteristic spectrum. If $-1/\mathrm{e} < a\tau < 0$, the two real branches give two real characteristic exponents in addition to the complex ones.

16.4 The final-size equation of an epidemic model

A common normalized SIR final-size equation is

$$\log\frac{s_\infty}{s_0} = -R_0(1 - s_\infty), \tag{131}$$

where $R_0 > 0$ and $0 < s_\infty \le s_0 \le 1$.

Theorem 16.4. *Assume $R_0 > 0$. Every positive algebraic solution is*

$$s_\infty = -\frac{1}{R_0} W_k(-R_0 s_0 e^{-R_0}). \tag{132}$$

The physically admissible branch must also satisfy $s_\infty \le s_0$ and the assumptions under which (131) was derived.

Proof. Exponentiating (131) gives

$$s_\infty = s_0 e^{-R_0} e^{R_0 s_\infty}.$$

Thus

$$(-R_0 s_\infty) e^{-R_0 s_\infty} = -R_0 s_0 e^{-R_0},$$

which proves (132). □

When two real branches are available, one may be a mathematically valid but physically inadmissible root. For example, if $s_0 = 1$ and $R_0 > 1$, the identity

$$W_{-1}(-R_0 e^{-R_0}) = -R_0$$

gives $s_\infty = 1$, the no-outbreak root, while W_0 gives the smaller nontrivial root. Initial conditions and model interpretation decide which one is relevant.

16.5 Inverting the prime-number scale

The rough relation

$$n \approx \frac{x}{\log x}$$

arises when inverting the prime number theorem. For $n \ge e$, solving the equality exactly and selecting its large solution gives

$$x = -n W_{-1}(-1/n). \tag{133}$$

Proof. Put $y = \log x$, so $x = e^y$. The equation $x/\log x = n$ becomes

$$\frac{e^y}{y} = n,$$

or

$$(-y)e^{-y} = -\frac{1}{n}.$$

The large solution has $y > 1$, so $-y < -1$, which selects W_{-1}: $y = -W_{-1}(-1/n)$. Exponentiating and using $e^{-W(z)} = W(z)/z$ gives $x = -nW_{-1}(-1/n)$. □

The principal branch would give a small root near $x = 1$, not the large-x inverse relevant to primes. Formula (75) immediately yields

$$-nW_{-1}(-1/n) = n\left(\log n + \log\log n + \frac{\log\log n}{\log n} + \cdots\right),$$

illustrating why the nonprincipal branch naturally enters prime asymptotics. The archive paper by Visser develops this viewpoint for bounds on the n-th prime.

17 Series in a logarithmic argument

For positive arguments it is sometimes better to expand the Wright omega function

$$\omega(t) = W_0(\mathrm{e}^t)$$

than to expand $W_0(x)$ directly. Its defining equation

$$\omega + \log \omega = t$$

has no exponentially large quantity.

Theorem 17.1 (Expansion centered at $x = \mathrm{e}$)**.** *Let* $h = \log x - 1$. *For* $x > 0$ *sufficiently close to* e,

$$W_0(x) = 1 + \frac{h}{2} + \frac{h^2}{16} - \frac{h^3}{192} - \frac{h^4}{3072} + \frac{13h^5}{61440} + \mathcal{O}(h^6). \tag{134}$$

Proof. Write $\omega = 1 + s$ and $t = 1 + h$. The defining equation becomes

$$h = s + \log(1 + s) = 2s - \frac{1}{2}s^2 + \frac{1}{3}s^3 - \frac{1}{4}s^4 + \cdots .$$

The derivative of the right side at $s = 0$ is $2 \neq 0$, so it has an analytic local inverse. Substitute $s = c_1 h + c_2 h^2 + \cdots$ and compare coefficients. Successive comparison gives

$$c_1 = \frac{1}{2}, \quad c_2 = \frac{1}{16}, \quad c_3 = -\frac{1}{192}, \quad c_4 = -\frac{1}{3072}, \quad c_5 = \frac{13}{61440},$$

which proves (134). $\square$

This representation, discussed in the Schroeder-series material in the archive, is useful when an application supplies $\log x$ directly. Its nonprincipal real analogue is not an ordinary Taylor series at the common endpoint: if

$$\nu(\eta) = -W_{-1}(-\mathrm{e}^{-\eta}),$$

then $\nu - \log \nu = \eta$ and the derivative with respect to ν vanishes at $(\eta, \nu) = (1, 1)$. The correct local expansion is therefore the signed square-root series (59), not a Taylor series. This is another manifestation of the branch-point asymmetry.

18 A brief guide to the complex branches

A complete construction of the Riemann surface belongs to complex analysis, but the main ideas follow directly from the defining map.

18.1 Local branches and the critical value

The complex derivative of $f(w) = w\mathrm{e}^w$ is

$$f'(w) = \mathrm{e}^w(1 + w).$$

Thus $w = -1$ is the only finite critical point, and its critical value is $-1/\mathrm{e}$. At every other preimage w_*, the complex inverse function theorem gives a unique analytic local branch through w_*. Near $-1/\mathrm{e}$, the same calculation as in Section 9 gives the complex square-root coordinate

$$p = \sqrt{2(1 + \mathrm{e}z)}.$$

At $z = 0$, the principal branch is regular and satisfies $W_0(0) = 0$. The other branches escape to complex infinity as $z \to 0$, creating logarithmic branching.

18.2 Indexing and the all-branch asymptotic expansion

For nonzero z, choose a determination of $\operatorname{Log} z$ and define

$$L_{1,k} = \operatorname{Log} z + 2\pi \mathrm{i} k, \qquad L_{2,k} = \operatorname{Log} L_{1,k}.$$

For large $|k|$, a solution is close to $L_{1,k} - L_{2,k}$. The standard indexing W_k is chosen so that

$$W_k(z) = L_{1,k} - L_{2,k} + o(1) \qquad (|k| \to \infty), \tag{135}$$

with consistent continuation away from branch cuts.

Theorem 18.1 (Complex all-branch expansion)**.** *Whenever $|L_{1,k}| \to \infty$ while the logarithms are chosen consistently,*

$$W_k(z) \sim L_{1,k} - L_{2,k} + \sum_{n=1}^{\infty} \frac{\mathcal{P}_n(L_{2,k})}{L_{1,k}^n}, \tag{136}$$

with the same Stirling polynomials $\mathcal{P}_n$ as in (64).

Proof. The complex equation is

$$W + \operatorname{Log} W = L_{1,k}.$$

Set $W = L_{1,k} - L_{2,k} + v$. On a neighborhood where the logarithms are consistent, exactly the algebra of Theorem 10.2 gives

$$\tau = 1 - \mathrm{e}^{-v} + \sigma v, \qquad \sigma = 1/L_{1,k}, \quad \tau = L_{2,k}/L_{1,k}.$$

The Lagrange-inversion calculation and coefficient identity are algebraic and remain valid over $\mathbb{C}$. This proves (136). □

This proof also shows that, for sufficiently large $|k|$, there is a solution near $L_{1,k} - L_{2,k}$: the implicit function theorem solves for v when σ and τ are small. Hence a nonzero complex argument has infinitely many Lambert values.

18.3 Derivative and conjugation

Every regular complex branch satisfies

$$W_k'(z) = \frac{W_k(z)}{z[1 + W_k(z)]}.$$

For a positive real x, complex conjugation preserves $w\mathrm{e}^w = x$. Consistency with the asymptotic indexing gives

$$W_{-k}(x) = \overline{W_k(x)}.$$

Thus nonreal values occur in conjugate pairs.

Literature note 18.2. Under the standard DLMF convention, W_0 is analytic on $\mathbb{C} \setminus (-\infty, -1/\mathrm{e}]$; the other branches are organized with cuts associated with the negative real axis and the logarithmic point at zero. Exact boundary labels require care because approaching a cut from above or below can change the branch index. The foundational paper of Corless et al. and DLMF Section 4.13 give the complete sheet structure and the standard conventions.

19 What the supplied archive contributes

The archive is not treated as a collection of interchangeable authorities. The foundational and reference sources establish notation and standard formulas; later papers contribute targeted approximations, bounds, algorithms, and applications. The following map records how the material was used.

Source in the archive	Main contribution to this article
Corless, Gonnet, Hare, Jeffrey, and Knuth (1996) [5]	Modern branch notation; Maclaurin, branch-point, and all-branch asymptotic expansions; symbolic integration and numerical evaluation. This is the foundational modern reference.
NIST DLMF, Section 4.13 [7]	Canonical definitions, branch domains, coefficient conventions, asymptotic formula, and the principal-branch integral representations proved in Section 13.
Barry et al. (2000) [2]	Piecewise analytic approximations for both real branches, with special care at endpoints and with formulas designed as iteration starts.
Banwell (2000) [1]	Use of W_0 to obtain explicit current–voltage formulas in diode and bipolar-transistor circuits; see Theorem 16.2.
Goličnik (2012) [8]	Closed forms in biochemical kinetics, including integrated Michaelis–Menten equations; see Theorem 16.1.
Chatzigeorgiou (2013) [3]	Elementary bounds for $W_{-1}(-\mathrm{e}^{-1-u})$. A self-contained proof and a midpoint error estimate appear in Theorem 11.5 and corollary 11.6.
Corcino, Corcino, and Mező (2019) [4]	Regular, associated, and Euler continued fractions. The elementary Euler fraction and its complete partial-sum proof appear in Theorem 12.3.
Iacono and Boyd (2017) [10]	Global analytic approximations and iterative refinements for the principal real branch, as well as approximation-based inequalities.
Lőczi (2022) [13]	Guaranteed and arbitrary-precision evaluation on the full real domains, especially logarithmic Newton recursions and explicit error control.
Howard (2025) and the accompanying manuscript [9]	Schroeder-type and transformed series, including expansions in $\log x$, branch-point series, and comparisons of approximation regimes; see Section 17.
Lehtonen (2016) [12]	A survey of ecological and evolutionary models whose implicit equations are naturally inverted by Lambert W.
Visser (2018) [15]	Use of W_{-1} in explicit approximations and bounds for the n-th prime; see Equation (133).
ProductLog notebook and expository PDF/TXT files	Computer-algebra notation, numerical cross-checks, and a broad inventory of identities. These were used as navigational aids rather than as primary proof sources.

Published maximum-error claims for fitted approximations depend on their stated domains and numerical searches. Rather than reproduce every fitted constant, this article concentrates on formulas whose correctness and error behavior can be proved directly: Taylor and Puiseux remainders, asymptotic orders, two-sided bounds, residual certificates, and monotone Newton iteration. The supplied Python program independently checks representative numerical behavior at high precision.

20 Summary and a branch-safe workflow

The Lambert function is best understood not as one mysterious formula but as a family of inverse maps. The real-variable theory can be organized around five decisions.

Step 1. Count the real solutions before choosing a symbol. For $-1/\mathrm{e} < x < 0$, there are two values. The value in $(-1, 0)$ is $W_0(x)$; the value in $(-\infty, -1)$ is $W_{-1}(x)$.

Step 2. Use the coordinate adapted to the endpoint. Near $x = 0$, use the Maclaurin series for W_0. Near the common branch point, use the signed square-root coordinate $p = \pm\sqrt{2(1 + \mathrm{e}x)}$. At the unbounded ends, use logarithmic coordinates.

Step 3. Keep branch qualifications in inverse identities. The equation $W(w\mathrm{e}^w) = w$ is true only on the inverse interval belonging to the selected branch. This is exactly analogous to the restricted ranges used for inverse trigonometric functions.

Step 4. For numerical work, combine an approximation with a correction and a certificate. The branch-aware starts of Theorem 14.4, logarithmic Newton iteration, and the residual bounds of Theorem 14.1 form a complete real-variable procedure.

Step 5. Expect most algebraic identities to be branch-independent, but most inequalities to be branch-dependent. The derivative, antiderivative, contour, and asymptotic calculations begin from $w\mathrm{e}^w = x$ and therefore apply on every regular branch. Monotonicity, curvature, signs, and bounds depend on the range of the selected inverse.

The two real branches are more symmetric than is sometimes suggested. They share the same branch point, signed Puiseux series, logarithmic asymptotic coefficient polynomials, inverse-function calculus, and monotone logarithmic Newton map. Their essential differences are global: W_0 is increasing and concave on its real domain, whereas W_{-1} is decreasing, changes curvature once, and has a logarithmic singularity at zero. The exact pairing formulas of Section 4 make this balance particularly clear.

21 Complements from the parallel treatments

The three parallel treatments merged into this volume derived the same core theory — branches, derivatives, Maclaurin and Puiseux series, asymptotic polynomials, bounds, and Newton iteration — with identical constants throughout. This section preserves their layers that the spine above does not contain. Attributions name the source package; the four-way concordance is in Appendix G.

21.1 Power towers and iterated exponentials

The fixed-point equation of the infinite tower $t_1 = a$, $t_{n+1} = a^{t_n}$ was solved in Section 15; the monograph treatment settles *convergence* completely.

Theorem 21.1 (Exact real power-tower interval). *For $a > 0$, the sequence of finite towers converges if and only if*

$$\mathrm{e}^{-\mathrm{e}} \le a \le \mathrm{e}^{1/\mathrm{e}}, \tag{137}$$

and throughout this interval its limit is $h = -W_0(-\log a)/\log a$ (with $h = 1$ at $a = 1$). For $1 < a < \mathrm{e}^{1/\mathrm{e}}$, the second real branch W_{-1} gives another positive solution of $h = a^h$, which is repelling and is not the limit of the finite towers.

Proof (monograph treatment). At a fixed point of $g(x) = a^x$, $g'(h) = h \log a = \log h$, so local attraction requires $1/\mathrm{e} < h < \mathrm{e}$; mapping the endpoints through $a = h^{1/h}$ produces (137). For $1 < a \le \mathrm{e}^{1/\mathrm{e}}$, $h^{1/h}$ increases on $[1, \mathrm{e}]$, so there is a unique fixed point $h \in [1, \mathrm{e}]$, and since $a \le h$

and g is increasing with $g([a,h]) \subseteq [a,h]$, the towers increase to h (including the neutral endpoint $a = \mathrm{e}^{1/\mathrm{e}}$). For $\mathrm{e}^{-\mathrm{e}} \le a < 1$, g is decreasing: the odd and even subsequences are monotone with limits $p \le h \le q$ forming a two-cycle $p = g(q)$, $q = g(p)$. Writing $u = -\log p$, $v = -\log q$, a nontrivial two-cycle forces $A := -\log a = u\mathrm{e}^v = v\mathrm{e}^u$, hence, with $r = u/v > 1$,

$$A = \frac{r \log r}{r-1}\, r^{1/(r-1)},$$

whose logarithm is strictly increasing in $\log r$ (this reduces to $2\sinh(s/2) > s$) with infimum 1; so $A > \mathrm{e}$, contradicting $A \le \mathrm{e}$. Hence $p = q = h$, including the neutral endpoint $a = \mathrm{e}^{-\mathrm{e}}$. Outside the interval no convergence occurs: for $a > \mathrm{e}^{1/\mathrm{e}}$ there is no fixed point, and for $0 < a < \mathrm{e}^{-\mathrm{e}}$ the unique fixed point has $|g'(h)| > 1$ and the injectivity of g prevents the iterates from reaching it exactly. The W_{-1} fixed point $H > \mathrm{e}$ has $g'(H) = \log H > 1$. □

The companion equation $x^y = y^x$ (fourth treatment): for positive $x \ne 1$, it is equivalent to $(\log y)/y = (\log x)/x =: a$, so

$$y = -\frac{1}{a} W_k(-a), \tag{138}$$

one branch reproducing $y = x$ and, for $0 < a < 1/\mathrm{e}$, the other giving the nontrivial partner (e.g. $x = 2$, $y = 4$); the branches merge at $x = y = \mathrm{e}$.

21.2 Fixed-point, Schröder, and inverse-Taylor methods

Two evaluation methods complement the monotone Newton theory of Section 14.

Proposition 21.2 (Logarithmic fixed-point iteration)**.** *The iteration $w_{n+1} = \log(x/w_n)$ has fixed points exactly at the nonzero branch values, with multiplier $-1/W$ there. It is locally attracting exactly when $|W| > 1$. Thus the principal iteration attracts only for $x > \mathrm{e}$, while the lower one attracts throughout $-1/\mathrm{e} < x < 0$. Contraction fails at $W_0(\mathrm{e}) = 1$ and at the branch point. Where the iteration converges, its rate is only linear.*

Theorem 21.3 (Inverse-Taylor (Schröder-type) corrections; monograph treatment)**.** *Let $y \ne -1$ approximate a branch value, $z_0 = y\mathrm{e}^y$, and $\Delta = x - z_0$. Then*

$$W(x) = y + \frac{\mathrm{e}^{-y}}{1+y}\Delta - \frac{\mathrm{e}^{-2y}(y+2)}{2(1+y)^3}\Delta^2 + \frac{\mathrm{e}^{-3y}(2y^2+8y+9)}{6(1+y)^5}\Delta^3 + \mathcal{O}(\Delta^4), \tag{139}$$

the coefficients being $W^{(j)}(z_0)/j!$ from Theorem 5.5. The linear truncation is exactly Newton's method for $w\mathrm{e}^w = x$; truncation after Δ^m turns an error $\varepsilon = y - W(x)$ into $\mathcal{O}(\varepsilon^{m+1})$. The quadratic truncation costs little more than Newton and gives third-order local accuracy; the powers of $(1+y)^{-1}$ warn against poor seeds near the branch point.

Convenient global seeds for one or two corrections (monograph treatment): $y_0 = \log(1+x)$ for W_0, $x > 0$ (an upper bound by Theorem 11.2); $y_0 = x$ for W_0 on $(-1/\mathrm{e}, 0)$; $y_0 = -T - \log T$ with $T = \log(1/(-x))$ for W_{-1}; and the Puiseux or logarithmic expansions at the ends.

21.3 Structural identities and the branch-exchange involution

The fourth treatment records several exact structures worth keeping explicit.

- **Branch-exchange involution.** For $0 < t \le 1$, $\sigma(t) = -W_{-1}(-t\mathrm{e}^{-t}) \ge 1$ satisfies $t\mathrm{e}^{-t} = \sigma(t)\mathrm{e}^{-\sigma(t)}$, and $s \mapsto -W_0(-s\mathrm{e}^{-s})$ inverts it: the two real branches encode the two inverse sides of the unimodal map $q \mapsto q\mathrm{e}^{-q}$. (The exact pair parametrization of Section 4 is the same involution in the gap coordinate Δ.)

- **Scaling.** If $w = W_k(x)$ and $n \in \mathbb{N}_0$, then

$$(nw)\mathrm{e}^{nw} = \frac{nx^n}{w^{n-1}}, \qquad \text{so} \qquad nW_k(x) = W_\ell\!\left(\frac{nx^n}{W_k(x)^{n-1}}\right)$$

on the branch containing the value; similarly for complex α with consistent powers.
- **Fixed points of W.** $W_k(z) = z$ forces $z\mathrm{e}^z = z$, i.e. $z = 0$ or $z = 2\pi \mathrm{i} n$; on the real principal branch the only fixed point is 0.
- **Unwinding.** The frequently written identity $W_k(z) = \operatorname{Log} z - \operatorname{Log} W_k(z) + 2\pi \mathrm{i} m$ carries a mandatory integer m recording the logarithm discrepancy; on the real branches with absolute values, $W + \log|W| = \log|x|$ holds unambiguously.
- **Closed Puiseux coefficients.** Writing the branch-point relation as $u = q\,u/G(u)$ with $G(u) = u\sqrt{2\big(1 + (u-1)\mathrm{e}^u\big)/u^2}$, Lagrange inversion gives the coefficients of the unified series S of Theorem 9.1 in closed form,

$$a_n = \frac{1}{n}\,[t^{n-1}]\left(\frac{t}{G(t)}\right)^n. \tag{140}$$

- **Monodromy.** Circling once around $-1/\mathrm{e}$ changes $p = \sqrt{2(1 + \mathrm{e}z)}$ to $-p$ and interchanges the two local sheets — on the real side, exactly $W_0 \leftrightarrow W_{-1}$ (third treatment).

21.4 A transcendence consequence

Theorem 21.4 (Fourth treatment)**.** *If $z \neq 0$ is algebraic, every value $W_k(z)$ is transcendental.*

Proof. If $w = W_k(z)$ were algebraic (and nonzero, since $z \neq 0$), then Lindemann–Weierstrass makes e^w transcendental, contradicting $\mathrm{e}^w = z/w$ algebraic. ☐

In particular the omega constant $\Omega = W_0(1)$ is transcendental. Values at transcendental arguments may still be algebraic — e.g. $W_0(\mathrm{e}) = 1$ and every manufactured value $W_k(a\mathrm{e}^a) = a$.

21.5 A practitioner's toolkit

Three habits from the third treatment simplify calculations in which W appears inside a larger model.

- **Parameter sensitivity.** If $w(\theta) = W_k(q(\theta))$ on a fixed branch, then

$$\nabla_\theta w = \frac{w}{q(1+w)}\,\nabla_\theta q$$

away from $q = 0$, $w = -1$ (with the limiting value $w' = q'$ at a principal-branch zero): every parameter derivative inherits the branch-point factor $(1+w)^{-1}$.
- **Differentiate in w, not in x.** For differentiable H, $\frac{\mathrm{d}}{\mathrm{d}x}H(W(x)) = H'(W(x))\,\mathrm{e}^{-W(x)}/(1 + W(x))$; computing $\mathrm{d}H/\mathrm{d}w$ first usually avoids nested quotient rules.
- **Parametrize by w.** The substitution $x = w\mathrm{e}^w$ traces each real branch with no numerical inversion ($w \geq -1$ for W_0, $w \leq -1$ for W_{-1}); a conjectured inequality $A(x) \leq W_0(x) \leq B(x)$ becomes the inverse-free statement $A(w\mathrm{e}^w) \leq w \leq B(w\mathrm{e}^w)$.

Floating-point hazards (monograph and fourth treatments): near the branch point, never form $1 + \mathrm{e}x$ by naive subtraction when x is given as $-1/\mathrm{e} + h$ — compute $p = \sqrt{2\mathrm{e}h}$ from h directly or use fused operations; for inputs so extreme that x overflows or underflows, carry $\log|x|$ and use the logarithmic Newton map (114); and prefer the residual $w + \log|w| - \log|x|$ to $w\mathrm{e}^w - x$ as a stopping test.

21.6 Further worked applications

- **Optimal patch residence (marginal value theorem).** With gain $G(t) = A(1 - \mathrm{e}^{-bt})$ and travel time τ, the rate-maximizing residence time solves $\mathrm{e}^{bt} = 1 + b(t + \tau)$, and with $y = 1 + b(t + \tau) > 1$,
$$t_* = \frac{-W_{-1}(-\mathrm{e}^{-1-b\tau}) - 1 - b\tau}{b} : \tag{141}$$
the principal branch would give negative residence time — a clean case where omitting W_{-1} loses the only admissible solution [12].
- **Wien displacement.** Maximizing $x^5/(\mathrm{e}^x - 1)$ gives $(5-x)\mathrm{e}^x = 5$, so $x_* = 5+W_0(-5\mathrm{e}^{-5}) \approx 4.9651142317$; the lower branch returns the boundary root $x = 0$ via $W_{-1}(-5\mathrm{e}^{-5}) = -5$. (Compare Theorem 15.7 with $n = 5$.)
- **Fall time with linear drag.** From $s(t) = v_\infty[t - (1 - \mathrm{e}^{-\gamma t})/\gamma]$ and $A = 1 + \gamma s/v_\infty \geq 1$, $t = [A+W_0(-\mathrm{e}^{-A})]/\gamma$; the lower branch gives the negative-time root of the extended equation.
- **Schwarzschild tortoise coordinate.** Inverting $r_* = r + 2M \log(r/2M - 1)$ for $r > 2M$:
$$r = 2M \left[1 + W_0\big(\mathrm{e}^{r_*/(2M)-1}\big)\right], \tag{142}$$
with the Wright omega form avoiding overflow for large r_*.
- **Prime counting.** From $p_n > n \log n$, inverting $u \log u$ gives $\pi(x) < x/W_0(x) = \mathrm{e}^{W_0(x)}$ for $x \geq 2$, and $\pi(x) \sim x/W_0(x)$ restates the prime number theorem; the nth-prime scale inversion is (133).

Every application above reduces to one of three normal forms (fourth treatment): $(ax+b)\mathrm{e}^{cx+d} = r$, solved by W directly; $x + \log x = c$, natural for W_0/Wright omega; and $x - \log x = c$, the two-branch inversion of Equation (5). Recognizing the form matters more than any single formula.

21.7 Generalizations and open directions

The *r-Lambert function* solves $w\mathrm{e}^w + rw = z$ [14]; it has $W_r'(z) = \left[\mathrm{e}^{W_r}(1 + W_r) + r\right]^{-1}$, and its branch points are images of the critical points $\mathrm{e}^w(1 + w) + r = 0$, themselves expressible by ordinary W. More generally, equations $\mathrm{e}^x \prod_i(x - t_i)/\prod_j(x - s_j) = a$ define generalized Lambert functions; the practical diagnostic is that ordinary W applies exactly when all appearances of the unknown gather into a single affine factor times the exponential of an affine factor (Theorem 15.4), and otherwise a genuine generalization is required. Approachable further directions collected by the fourth treatment: minimax rational approximations after mapping each unbounded branch to a finite interval; interval arithmetic with sharp enclosures at the square-root point; the interlacing of continued-fraction poles with the branch cut; real-variable analogues of branch monodromy in parameter-dependent models where two physical roots merge; and comparisons of W, Wright omega, and r-Lambert formulations for stiff delay equations.

22 Role in the Fabius–Rvachev corpus

Within this repository the Lambert function is not an isolated special function: the lower real branch W_{-1} is the canonical coordinate of the endpoint theory of the Fabius–Rvachev system. The left-endpoint asymptotics of the geometric-uniform laws have leading scale $-\log^2(1/x)/(2\log a)$, and inverting the associated saddle equations produces exactly the coordinate $-W_{-1}(-c\,x)$ with base-dependent normalization c — the dyadic all-orders endpoint and inverse-endpoint expansions, their periodic Gamma–zeta phases, and the general-base transform-level theory live in the exponents-and-q-series volume (Parts VI–VII) and in the inverse-endpoint volume of this draft corpus, with the exact phase-locked interpolation algebra formalized in `LambertPhaseLockedRichardson.lean`. A new finite algebraic rewrite tranche identifies the complete-homogeneous factors in those exact moments with complete Bell polynomials. The division-free identity `bellComplete_completeHomogeneousBellInput` and its factorially normalized form `completeHomogeneousEvalOn_eq_factorialNormalize_completeBellPolynomial` are proved in `CompleteHomogeneousBell.lean`; the input at index $k+1$ is $k!$ times the $(k+1)$-st power sum. The specialization in `LambertPhaseLockedBell.lean` takes the alphabet $(\lambda+j)^{-1}$, $0 \le j \le r$: `completeHomogeneousEvalOn_shiftedReciprocal_eq_bell` rewrites its degree-n complete-homogeneous value, while `sum_shiftedReciprocalLagrangeWeight_mul_invPow_card_add_eq_bell` and `sum_shiftedReciprocalLagrangeWeight_mul_invPow_eq_bell` give the exact Bell forms of the residual moments at exponents $r+1+n$ and $r+s$ ($s \ge 1$), respectively. The direct normalized specialization is stated for a field with rational-algebra structure, and the weighted moment forms assume characteristic zero. Within that setting field inversion is total, so no positivity or nonzero-shift hypothesis is added.

On the analytic side, the downstream modules `FabiusLambertPhaseLockedPullback.lean`, `CompleteHomogeneousAsymptotics.lean`, `LambertReciprocalAsymptotics.lean`, and `FabiusLambertPhaseExtraction.lean` now combine the arbitrary-order endpoint remainder with the growing reciprocal row: for every fixed row order r and residual depth S, the periodic estimator minus its first S exact residuals differs from the periodic endpoint term by $\mathcal{O}(\lambda^{-r-1-S})$, and it converges to the corresponding periodic value along every integer phase ray. This is a finite Poincaré statement, and the Bell conversion is likewise an exact finite rewrite. Neither gives convergence of an infinite residual series, a multiplicative relative-error hierarchy, a derivative extractor, uniformity in growing row order or residual depth, or the outstanding sign control for those extensions. The two-term inversion $\log(1/G^{-1}(y)) = \sqrt{2L\log(1/y)} - \frac{1}{2}\log\log(1/y) + \mathcal{O}(1)$ proved there is precisely the leading behavior of $-W_{-1}$ in the logarithmic regime of Section 10, and the fixed-point/cocycle structure behind those expansions is the $v - \log v = u$ geometry of Equation (5). This volume supplies the function-level foundations — branches, expansions, bounds, and certified evaluation — on which those corpus layers stand.

A A complete proof of Lagrange–Buermann inversion

This appendix proves the coefficient theorem used in the Maclaurin and asymptotic sections. The proof is algebraic and therefore works for formal power series; no question of numerical convergence is involved.

A.1 Formal residues

For a Laurent series $A(t) = \sum_{j\in\mathbb{Z}} a_j t^j$, define

$$\operatorname*{Res}_{t=0} A(t)\,\mathrm{d}t = a_{-1}.$$

Two elementary rules will be used.

Lemma A.1 (Derivative and substitution rules)**.** *Let $A(t)$ be a formal Laurent series.*

(i) $\operatorname{Res} A'(t)\,\mathrm{d}t = 0$.

(ii) *If $y = y(x) = c_1 x + c_2 x^2 + \cdots$ with $c_1 \neq 0$, then*

$$\operatorname*{Res}_{x=0} A(y(x))y'(x)\,\mathrm{d}x = \operatorname*{Res}_{y=0} A(y)\,\mathrm{d}y. \tag{143}$$

Proof. For (i), the derivative of $a_j t^j$ is $j a_j t^{j-1}$. A term in t^{-1} could arise only from $j = 0$, but its coefficient is then zero.

For (ii), linearity reduces the proof to $A(y) = y^m$. If $m \neq -1$, then

$$y(x)^m y'(x) = \frac{1}{m+1}\frac{\mathrm{d}}{\mathrm{d}x} y(x)^{m+1},$$

whose residue is zero by part (i); the residue of $y^m\,\mathrm{d}y$ is also zero. If $m = -1$, write $y(x) = c_1 x[1 + xB(x)]$. Then

$$\frac{y'(x)}{y(x)} = \frac{1}{x} + \frac{\mathrm{d}}{\mathrm{d}x}\log[1 + xB(x)],$$

so its residue is 1, exactly the residue of $\mathrm{d}y/y$. This proves the substitution rule term by term. □

A.2 Proof of the inversion formula

Suppose

$$y = x\phi(y), \qquad \phi(0) \neq 0. \tag{144}$$

There is a unique formal solution $y(x) = c_1 x + c_2 x^2 + \cdots$: after substituting the two series, the coefficient of x^n determines c_n from the preceding coefficients, beginning with $c_1 = \phi(0)$.

Let H be a formal power series and $n \geq 1$. Coefficient extraction is a residue:

$$[x^n]H(y(x)) = \operatorname*{Res}_{x=0} \frac{H(y(x))}{x^{n+1}}\,\mathrm{d}x. \tag{145}$$

From (144),

$$x = \frac{y}{\phi(y)}, \qquad \mathrm{d}x = \frac{\phi(y) - y\phi'(y)}{\phi(y)^2}\,\mathrm{d}y.$$

The substitution rule of lemma A.1 therefore changes (145) into

$$[x^n]H(y(x)) = \operatorname*{Res}_{y=0}\left[\frac{H(y)\phi(y)^n}{y^{n+1}} - \frac{H(y)\phi(y)^{n-1}\phi'(y)}{y^n}\right]\mathrm{d}y. \tag{146}$$

Now differentiate

$$\frac{H(y)\phi(y)^n}{y^n}.$$

Its residue is zero, so

$$0 = \operatorname*{Res}_{y=0} \left[\frac{H'(y)\phi(y)^n}{y^n} + n\frac{H(y)\phi(y)^{n-1}\phi'(y)}{y^n} - n\frac{H(y)\phi(y)^n}{y^{n+1}} \right] \mathrm{d}y.$$

Rearranging and comparing with (146) gives

$$[x^n]H(y(x)) = \frac{1}{n} \operatorname*{Res}_{y=0} \frac{H'(y)\phi(y)^n}{y^n}\,\mathrm{d}y = \frac{1}{n}[y^{n-1}]H'(y)\phi(y)^n.$$

This is (39). Taking $H(y) = y^m$ gives

$$[x^n]y(x)^m = \frac{m}{n}[y^{n-m}]\phi(y)^n,$$

which is (40). Thus every form of Lagrange inversion used in the article has now been proved.

B Coefficient tables and exact recurrences

The formulas in the main text are designed to generate coefficients, not merely to list them. The following tables are convenient for checking hand calculations and computer implementations.

B.1 Maclaurin coefficients

Write

$$W_0(x) = \sum_{n\geq 1} c_n x^n, \qquad c_n = \frac{(-n)^{n-1}}{n!}.$$

The first ten coefficients are

n	c_n	n	c_n
1	1	6	$-54/5$
2	-1	7	$16807/720$
3	$3/2$	8	$-16384/315$
4	$-8/3$	9	$531441/4480$
5	$125/24$	10	$-156250/567$

The last entry is intentionally left in a moderately factored rational form; the closed formula is usually more informative than a decimal expansion.

B.2 Signed branch-point coefficients

With $p = \pm\sqrt{2(1+\mathrm{e}x)}$, write

$$W = -1 + \sum_{n\geq 1} a_n p^n.$$

The first nine coefficients, generated by (52)–(53), are

n	a_n	n	a_n
1	1	6	$-221/8505$
2	$-1/3$	7	$680863/43545600$
3	$11/72$	8	$-1963/204120$
4	$-43/540$	9	$226287557/37623398400$
5	$769/17280$		

The same table serves both branches; only the sign of p changes.

B.3 Derivative polynomials

The recurrence in Theorem 5.5 begins

$$\begin{aligned}
P_1(w) &= 1, \\
P_2(w) &= -(w+2), \\
P_3(w) &= 2w^2 + 8w + 9, \\
P_4(w) &= -(6w^3 + 36w^2 + 79w + 64), \\
P_5(w) &= 24w^4 + 192w^3 + 622w^2 + 974w + 625.
\end{aligned}$$

Consequently

$$W_k^{(n)}(x) = \frac{\mathrm{e}^{-nW_k(x)} P_n(W_k(x))}{[1 + W_k(x)]^{2n-1}}.$$

This form avoids differentiating an increasingly complicated quotient from scratch.

C Numerical experiments and reproducibility

The archive accompanying this article contains `numerical_experiments.py`. It uses `mpmath` at 80 decimal digits to obtain reference values and then implements, independently, the approximations proved in the article:

- the signed branch-point series for both real branches;
- the logarithmic asymptotic expansion for W_0 and W_{-1};
- the three Padé approximants of Section 12;
- the two-sided W_{-1} bounds;
- the branch-safe logarithmic Newton iteration.

From the directory containing the source, run

```
python3 numerical_experiments.py --output-dir figures
```

The required packages are `mpmath`, numpy, and `matplotlib`; the script does not use network access. Every plotting function writes a vector PDF, so the figures remain sharp when enlarged. The reference values and defining-equation residuals are also written to `numerical_values.txt`.

Listing 1: High-precision reference values generated by the supplied script.

```
Reference values generated with mpmath at 80 decimal digits
x branch W_k(x) |W exp(W)-x|
--------------------------------------------------------------------------------
-1/e 0 -1.0 0.0
-0.35 0 -0.71663881645607385059 5.2711e-82
-0.35 -1 -1.3497172521922488334 5.2711e-82
-0.10 0 -0.11183255915896296483 0.0
-0.10 -1 -3.5771520639572972184 1.3178e-82
-0.001 -1 -9.1180064704027401213 1.2354e-83
0 0 0.0 0.0
0.1 0 0.091276527160862264 3 1.3178e-82
1 0 0.567143290409783873 1.0542e-81
e 0 1.0 0.0
10 0 1.7455280027406993831 3.3735e-80
10^6 0 11.383358086140052622 1.1054e-74
```

The reference implementation is not used as a hidden proof. Its purpose is to detect transcription mistakes, illustrate the proved convergence orders, and make every graph reproducible. Analytic statements in the main text are established independently of the plots.

D Problems with complete solutions

These exercises review the main ideas without requiring complex analysis.

Problems

P1. Prove that the equation $we^w = -1/10$ has exactly two real solutions, and locate each one in an interval of length 1.

P2. Let $a \geq -1$. Prove $W_0(ae^a) = a$. Let $a \leq -1$. Prove $W_{-1}(ae^a) = a$. Explain why the branch restrictions cannot be removed.

P3. Solve $x + 2\log x = 5$ for $x > 0$.

P4. Evaluate $\int_{-1/e}^{0} W_{-1}(x)\,dx$.

P5. Show directly from the defining equation that

$$\frac{d}{dx}e^{W_0(x)} = \frac{1}{1+W_0(x)} \qquad (x > -1/e).$$

P6. For $-1/e < x < 0$, prove $W_0(x) + W_{-1}(x) < -2$ without differentiating either branch.

P7. Determine the number of positive real solutions of $x^x = A$ for $A > 0$, and express them with the correct Lambert branches.

P8. Derive the first three nonconstant terms in the expansion of $W_0(x)$ at $x = 0$ by substituting $W_0(x) = a_1x + a_2x^2 + a_3x^3 + \mathcal{O}(x^4)$ into the defining equation.

P9. Let $x > e$, $L = \log x$. Prove that $L - \log L < W_0(x) < L$.

P10. Suppose an approximation w to a selected real Lambert value satisfies $w \neq 0, -1$. Derive one logarithmic Newton step without using e^w.

Solutions

S1. Since $-1/e < -1/10 < 0$, Theorem 2.1 gives exactly two solutions. On $(-1, 0)$, $(-1)e^{-1} < -1/10 < 0$, so the principal solution lies in $(-1, 0)$. For the lower branch,

$$-4e^{-4} \approx -0.0733 > -0.1, \qquad -3e^{-3} \approx -0.1494 < -0.1.$$

Because we^w is decreasing on $(-\infty, -1)$, the second solution lies in $(-4, -3)$.

S2. The map $w \mapsto we^w$ is one-to-one from $[-1, \infty)$ onto $[-1/e, \infty)$, so its inverse there sends ae^a back to a. This is W_0. The same argument on $(-\infty, -1]$, where the map is one-to-one onto $[-1/e, 0)$, gives the W_{-1} identity. Without the range restriction, an input in $(-1/e, 0)$ has two preimages, so a single inverse formula cannot select both.

S3. Divide by 2: $x/2 + \log x = 5/2$. Exponentiation gives $xe^{x/2} = e^{5/2}$. Hence

$$\frac{x}{2}e^{x/2} = \frac{1}{2}e^{5/2}, \qquad x = 2W_0\left(\frac{1}{2}e^{5/2}\right).$$

The argument is positive, so only W_0 is real.

S4. By (30), the answer is $-3/e$. Directly, put $w = W_{-1}(x)$. Then w runs from -1 to $-\infty$, and

$$\int we^w(1+w)\,dw = \left[e^w(w^2 - w + 1)\right]_{-1}^{-\infty} = -\frac{3}{e}.$$

S5. Differentiate and use (20):

$$\frac{d}{dx}e^{W_0(x)} = e^{W_0(x)}W_0'(x) = e^{W_0(x)}\frac{e^{-W_0(x)}}{1+W_0(x)}.$$

The factors cancel.

S6. Let $\Delta = W_0(x) - W_{-1}(x) > 0$. By (13), the sum is $-\Delta\coth(\Delta/2)$. Put $y = \Delta/2$. The function $y\cosh y - \sinh y$ has value 0 at 0 and positive derivative $y\sinh y$, so $y\coth y > 1$. The sum is therefore less than -2.

S7. Put $u = \log x$. Then $x^x = A$ becomes $ue^u = \log A$, so

$$x = \exp[W_k(\log A)].$$

If $A > 1$, then $\log A > 0$ and there is one solution, from W_0. If $A = 1$, the positive solution is $x = 1$. If $\mathrm{e}^{-1/\mathrm{e}} < A < 1$, then $-1/\mathrm{e} < \log A < 0$ and there are two solutions, from W_0 and W_{-1}. If $A = \mathrm{e}^{-1/\mathrm{e}}$, the two coalesce at $x = \mathrm{e}^{-1}$. If $0 < A < \mathrm{e}^{-1/\mathrm{e}}$, there is no positive real solution.

S8. Expand $\mathrm{e}^W = 1 + W + W^2/2 + \mathcal{O}(x^3)$. Then

$$W\mathrm{e}^W = W + W^2 + \frac{1}{2}W^3 + \mathcal{O}(x^4).$$

Substitution gives

$$a_1x + (a_2 + a_1^2)x^2 + (a_3 + 2a_1a_2 + a_1^3/2)x^3 + \mathcal{O}(x^4) = x.$$

Thus $a_1 = 1$, $a_2 = -1$, and $a_3 = 3/2$.

S9. Since $L\mathrm{e}^L = Lx > x$ and $w\mathrm{e}^w$ is increasing for $w > 0$, one has $W_0(x) < L$. Put $a = L - \log L$. Then

$$a\mathrm{e}^a = x\left(1 - \frac{\log L}{L}\right) < x,$$

so the same monotonicity gives $a < W_0(x)$.

S10. Apply Newton's method to $G(w) = w + \log|w| - \log|x|$. Since $G'(w) = 1 + 1/w$,

$$\begin{aligned} w^+ &= w - \frac{w + \log|w| - \log|x|}{1 + 1/w} \\ &= \frac{w}{1+w}\left(1 + \log\left|\frac{x}{w}\right|\right). \end{aligned}$$

This is (108).

E Compact formula sheet

Topic	Formula and branch condition
Definition	$W_k(x)\mathrm{e}^{W_k(x)} = x$.
Real domains	$W_0 : [-1/\mathrm{e}, \infty) \to [-1, \infty)$, increasing; $W_{-1} : [-1/\mathrm{e}, 0) \to (-\infty, -1]$, decreasing.
Derivative	On W_0 for $x > -1/\mathrm{e}$, and on W_{-1} for $-1/\mathrm{e} < x < 0$, $W_k'(x) = \mathrm{e}^{-W_k(x)}/[1 + W_k(x)]$. When $x \neq 0$, this is also $W_k(x)/\{x[1 + W_k(x)]\}$; separately, $W_0'(0) = 1$.
Second derivative	On the same smooth domains, $W_k''(x) = -\mathrm{e}^{-2W_k(x)}[W_k(x)+2]/[1+W_k(x)]^3$. When $x \neq 0$, this is also $-W_k(x)^2[W_k(x) + 2]/\{x^2[1 + W_k(x)]^3\}$; separately, $W_0''(0) = -2$.
Antiderivative	$\int W_k(x)\,\mathrm{d}x = \mathrm{e}^{W_k(x)}[W_k(x)^2 - W_k(x) + 1] + C$.
Maclaurin	$W_0(x) = \sum_{n\geq1}(-n)^{n-1}x^n/n!$, $\lvert x\rvert < 1/\mathrm{e}$.

Topic	Formula and branch condition
Branch point	$W_{0,-1}(x) = -1 + p - p^2/3 + 11p^3/72 - \cdots$, $p = \pm\sqrt{2(1+\mathrm{e}x)}$.
Large W_0	$W_0(x) = L_1 - L_2 + L_2/L_1 + \cdots$, $L_1 = \log x$, $L_2 = \log L_1$.
Small-negative W_{-1}	Same coefficient formula with $L_1 = \log(-x) < 0$, $L_2 = \log[-L_1]$.
Exact branch pair	If $\Delta = W_0 - W_{-1}$, then $W_0 = -\Delta/(\mathrm{e}^{\Delta} - 1)$ and $W_{-1} = -\Delta/(1 - \mathrm{e}^{-\Delta})$.
Logarithmic Newton	$w^+ = \dfrac{w}{1+w}\,(1 + \log\lvert x/w\rvert)$.
Residual certificate	$\lvert w - W_k(x)\rvert \le \lvert w\mathrm{e}^w - x\rvert / \min_I \lvert \mathrm{e}^t(1+t)\rvert$, provided the interval I between w and the root avoids -1.

F Lean formalization crosswalk

The repository's Lean library carries a two-branch development of the real Lambert function, and several of the foundational statements of this guide are exact theorems there. This appendix records the correspondence, in the discipline of the other frontier volumes: a row names declarations, all in namespace `Fabius`, and a row is listed only when the Lean statement has the displayed mathematical content. Where the Lean statement is weaker, more general, or differently normalized than the guide's, the difference is stated. Everything not listed here is unformalized.

The two branches are `principalLambertW` (W_0, totalized, intended domain $[-1/\mathrm{e}, \infty)$; module `PrincipalLambertW`) and `lowerLambertW` (W_{-1}, intended domain $[-1/\mathrm{e}, 0)$; module `LowerLambertW`). Both are defined as the logarithm of an inverse of $u \mapsto u \log u$ on the appropriate ray, so "$W(x)$" is a total real function whose values off the intended domain are junk; every theorem below carries the domain hypothesis explicitly.

The focused branch-pair and Bernoulli tranche is exhaustive at this checkpoint: `LambertWBranchPairing` has $0 + 7$ public definitions/theorems, `LambertWGapBijection` has $4 + 16$, and `LambertWBranchSymmetry` has $0 + 9$, while `LambertWBranchGapBernoulli` has $0 + 5$. Their four-module union is therefore four definitions and 37 theorems, 41 public declarations in all. The preceding exact-radius checkpoint had the then-current census 903/11,447; the complex value-completion theorem brought the next historical checkpoint to 903/11,448. The local five-declaration fixed-column extension of `QBinomialTheoremInfinite.lean` separately recorded the historical 903/11,453 checkpoint, while its first union with the parallel algebraic moment-polynomial branch recorded 904/11,462. The later sibling `GeometricUniformMomentPolynomial.lean` module has $1 + 8$ public definitions/theorems and raised the next historical census to 904/11,457. Its companion `GeometricUniformMomentPolynomialBridge.lean` has the exhaustive count $0 + 1$ and raised the next historical census to 905/11,458. Its sole theorem, `geometricUniformMomentPolynomial_eval₂_eq_mgf_taylorCoefficient`, supplies the real-$|q| < 1$ actual-MGF normalization and makes `p7:eq:Pn-def` Exact in that regime. It supplies no complex-q infinite-product realization by itself. The subsequent exhaustive $1 + 2$ `GeometricUniformComplexMomentProduct.lean` leaf consists of `geometricUniformComplexMomentProduct`, `hasProdLocallyUniformly_geometricUniformComplexMomentProduct`, and `geometricUniformMomentPolynomial_eval₂_eq_complexMomentProduct_taylorCoefficient`. For every complex $\|q\| < 1$, these give the actual locally uniform product and identify its normalized Taylor coefficient with the recursive polynomial. This is a complex analytic analogue, not a complex probability-moment extension of `p7:eq:Pn-def`, whose Exact status remains restricted to real $|q| < 1$. At this historical point the canonical q-monograph compound `thm:qF-moment-polynomial` remained Partial because neither the global `RatFunc` identification nor its pole-clearing polynomial continuation at roots was formalized. The complex leaf raised the next incoming-branch checkpoint to the historical 906/11,461, and in the merged chronology it formed the historical 918/11,568 checkpoint. The exhaustive zero-definition/one-theorem `HalfQBinomialRootSimplicity.lean` leaf then gave the merged-main 919/11,569 checkpoint. Its `halfQBinomial_sum_rootMultiplicity_two_pow`, together with the existing complete rational root classification, makes `cor:halfbase-root-locus` Exact. No arbitrary-base or arbitrary-field simplicity is claimed, and `cor:qbinom-inversion-law` remains Partial. The subsequent exhaustive $1 + 2$ `GeometricUniformExteriorComplexMomentGerm.lean` sibling consists of `geometricUniformExteriorComplexMomentGerm`, `analyticAt_geometricUniformExteriorComplexMomentGerm`, and `geometricUniformMomentPolynomial_eval₂_eq_exteriorComplexMomentGerm_taylorCoefficient`. For every complex q with $1 < \|q\|$, these define the actual reciprocal germ, prove its analyticity at the origin, and identify its normalized Taylor coefficient with the recursive polynomial. No unit-circle or global rational-function claim is made. The exterior leaf raises the census to 920/11,572.

The final exhaustive $0 + 3$ `GeometricUniformMomentPolynomialDegree.lean` sibling consists of `coeff_geometricUniformMomentPolynomial_choose_two`, `coeff_geometricUniformMomentPolynomial_choose_two_sub_one`, and `geometricUniformMomentPolynomial_natDegree_eq`. With $T_n = \binom{n}{2}$, these prove the top coefficient $B_n(1)/n!$, the subleading coefficient $-B_n(1)/n! + B_{n-1}(1)/(2(n-1)!)$ for $n \geq 2$, and degree T_n for even n or $n = 1$, one less otherwise. Here Lean's `bernoulli'` is $B_n(1) = (-1)^n B_n$ under Mathlib's convention $B_1 = -1/2$. This makes `p7:thm:Pn` and `prop:qF-P-degree-sharp` Exact. At that checkpoint the global `thm:qF-moment-polynomial` was still Partial for the separate `RatFunc`/root-continuation gap. The sharp leaf raises the census to the historical 921/11,575 checkpoint. The exhaustive `RvachevLaurentLeading.lean` $(1+6)$ sibling then gives 922/11,582 and makes `is:p2:thm:Laurent-leading` Exact. The exhaustive `FinitePrefixAppellRecovery.lean` $(11 + 17)$ sibling gives the 923/11,610 checkpoint and makes `is:p2:thm:finite-prefix-expansion` and `is:p2:thm:exact-recovery` Exact as finite rational identities.

The final exhaustive $1 + 4$ `GeometricUniformMomentRatFunc.lean` sibling consists of `geometricUniformMomentRatFunc`, `qFactorial_mul_geometricUniformMomentRatFunc`, `eval_geometricUniformMomentRatFunc_eq_complexMomentProduct_taylorCoefficient`, `eval_geometricUniformMomentRatFunc_eq_exteriorComplexMomentGerm_taylorCoefficient`, and `eval_geometricUniformMomentRatFunc_one`. It packages one `RatFunc` over the rationals, proves the global q-factorial clearing identity, identifies that same rational function with both strict off-unit-circle Taylor-coefficient regimes, and handles $q = 1$ through $[n]_1! = n!$. Together with the recursive polynomial leaf, this makes `thm:qF-moment-polynomial` Exact and gives the historical 924/11,615 checkpoint. No analytic value, pole order, or divisor is asserted at a genuine unit-root pole. The later `weightedSumDistribution_real_Ici_eq_rvachevUp_of_nonneg` and `integral_pow_weightedSumDistribution_eq_mul_intervalIntegral_rvachevUp` theorems make `prop:up-tail` and `cor:up-moments` Exact. The unrelated exhaustive (1+1) `RvachevLegendreBiorthogonality.lean` leaf consists of `rvachevLegendreAnalysisKernel` and `rvachevLegendreBiorthogonality`; it closes the translated-kernel definition and finite Legendre biorthogonality, but not the larger Fourier–Bessel or projector claims. These additions give the incoming-branch 925-module/11,619-declaration census. After the two shared fixed-column names are counted once, the other three local declarations give the merged live 925-module/11,622-declaration census. The q forward ledger is 181 Exact / 78 Partial / 15 None / 8 interface rows, and its source concordance is 94 Lean-proved / 384 human-proved frontier / 60 non-applicable / 9 conjectures. The pre-merge Guide PDF receipt remains recorded solely as historical provenance and is not the render receipt for this merged edition.

Statement in this guide	Exact Lean declaration(s)	Remarks
Theorem 2.1: W_0 is a strictly increasing bijection $[-1/\mathrm{e}, \infty) \to [-1, \infty)$, W_{-1} a strictly decreasing bijection $[-1/\mathrm{e}, 0) \to (-\infty, -1]$, and every real solution of $w\mathrm{e}^w = x$ is one of the two	`principalLambertW_strictMonoOn`, `principalLambertW_image_Ioi`, `principalLambertW_image_Icc`, `neg_one_le_principalLambertW`, `lowerLambertW_strictAntiOn`, `lowerLambertW_image`, `lowerLambertW_le_neg_one`, `eq_principalLambertW_or_eq_lowerLambertW`	The last, in module `LambertBranchDichotomy`, is the dichotomy: $w\mathrm{e}^w = z$ forces $w = W_0(z)$ or $w = W_{-1}(z)$. The images are stated for the open ray $(-1/\mathrm{e}, \infty)$ and the interval $[-1/\mathrm{e}, 0]$ on the principal side and the open interval $(-1/\mathrm{e}, 0)$ on the lower side; the endpoint values are separate lemmas.
Corollary 2.4: $W_0(a\mathrm{e}^a) = a$ for $a \geq -1$ and $W_{-1}(a\mathrm{e}^a) = a$ for $a \leq -1$	`principalLambertW_unique`, `lowerLambertW_unique_of_mem_Ico`; the defining equations $W(x)\mathrm{e}^{W(x)} = x$ are `principalLambertW_mul_exp` and `lowerLambertW_mul_exp_of_mem_Ico`	Stated as uniqueness: any $w \geq -1$ with $w\mathrm{e}^w = z$ equals $W_0(z)$, and dually for $w \leq -1$.
Theorem 5.1, first and second derivatives	`deriv_principalLambertW`, `deriv_deriv_principalLambertW` (module `LambertWCurvature`); `deriv_lowerLambertW`, `deriv_deriv_lowerLambertW`	The principal-branch first derivative is in the form (20), the lower-branch one in the form (22); both second derivatives are in the exponential form (21). $W_0'(0) = 1$ is `principalLambertW_isEquivalent_zero`, stated as $W_0(z) \sim z$ at 0. Positivity of W_0' is `deriv_principalLambertW_pos`.

Statement in this guide	Exact Lean declaration(s)	Remarks
Theorem 5.4: W_0 strictly concave; W_{-1} strictly convex on $(-1/\mathrm{e}, -2/\mathrm{e}^2)$, inflection at $-2/\mathrm{e}^2$, strictly concave on $(-2/\mathrm{e}^2, 0)$	`strictConcaveOn_principalLambertW`, `strictConvexOn_lowerLambertW_left`, `strictConcaveOn_lowerLambertW_right`, `deriv_deriv_lowerLambertW_eq_zero_iff`	Lean's concavity of W_0 is on the closed ray $[-1/\mathrm{e}, \infty)$, slightly more than stated here; the lower-branch pieces are on the closed interval $[-1/\mathrm{e}, -2/\mathrm{e}^2]$ and the half-open $[-2/\mathrm{e}^2, 0)$. The inflection is stated as $W''_{-1}(x) = 0 \iff x = -2/\mathrm{e}^2$ on the open domain.
Theorem 8.2, Theorem 8.5: the Maclaurin coefficients $(-n)^{n-1}/n!$	`coeff_lambertW` (module `LambertWSeries`)	A statement about the *formal* series over $\mathbb{Q}$: the compositional inverse of $X\mathrm{e}^X$ has $(n+1)$-st coefficient $(-(n+1))^n/(n+1)!$. It is obtained from the library's Lagrange–Bürmann inversion (Theorem 8.1), and carries no statement about convergence, the radius $1/\mathrm{e}$, or the endpoint; those remain unformalized.
Theorem 9.1, leading term: $W_0(x) + 1 \sim q$ and $W_{-1}(x) + 1 \sim -q$ as $x \downarrow -1/\mathrm{e}$, with $q = \sqrt{2(1 + \mathrm{e}x)}$	`principalLambertW_add_one_isEquivalent_branchPoint`, `lowerLambertW_add_one_isEquivalent_branchPoint` (module `LambertWBranchPointAsymptotics`); non-differentiability at the branch point, `principalLambertW_not_differentiableAt_branchPoint` and `lowerLambertW_not_differentiableAt_branchPoint` (module `LambertWBranchPointGeometry`)	Only the leading square-root term is formal; the full signed Puiseux series and its coefficients are not.
Lemma 11.1: $t/(1+t) \le \log(1+t) \le t$ for $t > -1$	Mathlib: `Real.log_le_sub_one_of_pos` and `Real.one_sub_inv_le_log_of_pos`	Both halves are Mathlib lemmas (not repository declarations), stated at $1 + t > 0$; the strict upper form is `Real.log_lt_sub_one_of_pos`.

Statement in this guide	Exact Lean declaration(s)	Remarks
Theorem 11.2: $x/(1+x) \le W_0(x) \le \log(1+x) \le x$ for $x \ge 0$, strict for $x > 0$	`principalLambertW_elementary_bounds`, `div_one_add_le_principalLambertW`, `principalLambertW_le_log_one_add`, `principalLambertW_le_self`, and the strict `div_one_add_lt_principalLambertW`, `principalLambertW_lt_log_one_add` (module `LambertWElementaryBounds`)	Proved from a single inequality with no Lambert content, valid for *every* real w: `exp_le_one_add_mul_exp_self`, $\mathrm{e}^w \le 1+w\mathrm{e}^w$, strict for $w \neq 0$; the guide's $w = W_0(x)$ is one instance, and `le_log_one_add_mul_exp` states the upper bound for every real w in the form $w \le \log(1+w\mathrm{e}^w)$.
Theorem 11.7: for $-1/\mathrm{e} < x < 0$ and $\eta = \log(1/(-x))$, $-\eta - \eta\log\eta/(\eta-1) \le W_{-1}(x) < -\eta - \log\eta$	`lowerLambertW_near_zero_bounds`, with the halves `neg_log_sub_div_le_lowerLambertW` and `lowerLambertW_lt_neg_log_sub` (module `LowerLambertW`)	The mechanism is the bracket for the inverse of $z - \log z$, stated with no Lambert content for every $y > 1$ with $y - \log y = \eta$: `SubLog.bracket_of_sub_log_eq`, from `SubLog.strictMonoOn_sub_log`. The reduction $y = -W_{-1}(x), y - \log y = \eta$ is `neg_lowerLambertW_sub_log_eq`, and $\eta > 1$ on the whole domain is `one_lt_log_one_div_neg`, so the denominator $\eta - 1$ is never a side condition for the caller.

Statement in this guide	Exact Lean declaration(s)	Remarks
Theorem 5.5: every derivative of either branch is $W^{(n)} = \mathrm{e}^{-nW} P_n(W)/(1+W)^{2n-1}$ with $P_{n+1} = (1+w)P_n' - (nw+3n-1)P_n$	`lambertDerivPoly`, `lambertDerivPoly_succ`, `iteratedDeriv_principalLambertW`, `iteratedDeriv_lowerLambertW` (module `LambertWHigherDerivatives`); the general engine is `AutonomousODE.iteratedDeriv_eq_comp` (module `AutonomousIteratedDeriv`)	Lean's P_n is this guide's P_{n+1}: the recurrence is stated as $P_{n+1} = (1+w)P_n' - ((n+1)w+3n+2)P_n$ from $P_0 = 1$, and the formula as $W^{(n+1)} = \frac{\mathrm{e}^{-(n+1)W} P_n(W)}{(1+W)^{2n+1}}$, on the open domains $(-1/\mathrm{e}, \infty)$ and $(-1/\mathrm{e}, 0)$. The engine is stated for an arbitrary solution of an autonomous equation $W' = \varphi \circ W$ on an open set, with the derivative family as hypotheses, so it applies beyond Lambert's function.
Theorem 6.1: $\int W_k = \mathrm{e}^{W_k}(W_k^2 - W_k + 1) = x(W_k - 1 + 1/W_k)$ and $\int W_k/x = W_k + \frac{1}{2}W_k^2$	`hasDerivAt_principalLambertW_antiderivative`, `hasDerivAt_lowerLambertW_antiderivative`, `hasDerivAt_principalLambertW_add_half_sq`, `hasDerivAt_lowerLambertW_add_half_sq` (module `LambertWAntiderivative`); the general forms `LambertAntiderivative.hasDerivAt_exp_mul_sq_sub_add_one`, `LambertAntiderivative.hasDerivAt_add_half_sq` and the two-form identity `LambertAntiderivative.exp_mul_sq_sub_add_one_eq`	Stated as derivative facts ($F' = W$, $G' = W/x$) on the open branch domains, which is the content of the antiderivative statement. The general forms assume only $W' = (\mathrm{e}^W(1+W))^{-1}$ at the point, with $W \neq -1$, so they hold for any local inverse of $w \mapsto w\mathrm{e}^w$; the x-form needs $W \neq 0$ and the W/x form needs $x \neq 0$, as in the guide.

Statement in this guide	Exact Lean declaration(s)	Remarks
Theorem 11.3: for $x > \mathrm{e}$, $L = \log x$, $L - \log L < W_0(x) < L - \log(L - \log L) < L$	`principalLambertW_log_bounds`, with the halves `log_sub_log_log_lt_principalLambertW`, `principalLambertW_lt_log_sub_log_log_sub_log_log`, `principalLambertW_lt_log` (module `LambertWLogBounds`); the non-strict outer bracket for $x \geq \mathrm{e}$ is `principalLambertW_log_bracket` (module `PrincipalLambertWAtTop`) and the non-strict sharper upper bound `principalLambertW_le_log_sub_log_log_sub_log_log`	Exact. All from $W + \log W = L$ (`principalLambertW_add_log`) and monotonicity of the logarithm; the equality $W_0(\mathrm{e}) = 1$ at the endpoint is `principalLambertW_exp_one`.
Proposition 3.1: $(a+b)\mathrm{e}^{a+b} = xy(1/a+1/b)$ for Lambert values $a, b \neq 0$ of x, y, hence $a + b = W_m(xy(1/a + 1/b))$ on the branch containing $a+b$	`mul_exp_add_of_mul_exp` (the identity, for any reals satisfying their defining equations), `add_eq_principalLambertW_of_mul_exp`, `add_eq_lowerLambertW_of_mul_exp`, `principalLambertW_add_principalLambertW` (module `LambertWAdditionIdentity`)	Exact. The branch consequence is stated through the uniqueness theorems: $a+b \geq -1$ selects W_0, and $a+b \leq -1$ with the product in $[-1/\mathrm{e}, 0)$ selects W_{-1}; for $x, y > 0$ the two principal values add to the principal value, with no side condition.
Theorem 4.1, forward direction: with $\Delta = W_0(x) - W_{-1}(x) > 0$ on $(-1/\mathrm{e}, 0)$, $W_0 = -\Delta/(\mathrm{e}^{\Delta} - 1)$, $W_{-1} = -\Delta/(1 - \mathrm{e}^{-\Delta}) = -\Delta\mathrm{e}^{\Delta}/(\mathrm{e}^{\Delta} - 1)$, $x = W_0\mathrm{e}^{W_0}$ in the Δ-form, and the $t = \mathrm{e}^{\Delta}$ forms	`principalLambertW_sub_lowerLambertW_pos`, `lowerLambertW_eq_principalLambertW_mul_exp_gap`, `principalLambertW_eq_neg_gap_div`, `lowerLambertW_eq_neg_gap_mul_exp_div`, `lowerLambertW_eq_neg_gap_div_one_sub_exp_neg`, `eq_neg_gap_div_mul_exp`, `principalLambertW_lowerLambertW_eq_of_exp_gap` (module `LambertWBranchPairing`)	Exact for the forward direction. The whole computation is $W_{-1} = W_0\mathrm{e}^{\Delta}$ from dividing the two defining equations.

Statement in this guide	Exact Lean declaration(s)	Remarks
Theorem 4.1, converse: every $\Delta > 0$ inserted into the formulas produces one and only one $x \in (-1/\mathrm{e}, 0)$, with the displayed values equal to $W_0(x)$ and $W_{-1}(x)$	`gapPrincipal`, `gapLower`, `gapArg` (the three formulas as functions of Δ), `branchGap` (the map $x \mapsto W_0(x) - W_{-1}(x)$); `gapArg_mem_Ioo`, `principalLambertW_gapArg`, `lowerLambertW_gapArg`, `branchGap_gapArg`, `gapArg_branchGap`; the two directions packaged as `branchGap_invOn` and `branchGap_bijOn`; and the three $t > 1$ coordinate formulas are `principalLambertW_gapArg_log`, `lowerLambertW_gapArg_log`, and `gapArg_log` (module `LambertWGapBijection`)	Exact, and stated as a bijection: $x \mapsto \Delta$ is a bijection from $(-1/\mathrm{e}, 0)$ onto $(0, \infty)$ with the explicit inverse $\Delta \mapsto a(\Delta)\mathrm{e}^{a(\Delta)}$, $a(\Delta) = -\Delta/(\mathrm{e}^{\Delta} - 1)$. The proof is uniqueness on each branch: $b(\Delta) = a(\Delta)\mathrm{e}^{\Delta}$ equals $a(\Delta) - \Delta$, so $b\mathrm{e}^{b} = a\mathrm{e}^{a}$, and $-1 < a < 0$, $b < -1$ are the tangent-line inequalities $\Delta < \mathrm{e}^{\Delta} - 1$ and $\mathrm{e}^{\Delta}(1 - \Delta) < 1$.
Corollary 4.2: the exact quotient, sum, and product identities, and $W_0(x) + W_{-1}(x) < -2$, $0 < W_0(x)W_{-1}(x) < 1$ on $(-1/\mathrm{e}, 0)$	`lowerLambertW_div_principalLambertW_eq_exp_branchGap`, `principalLambertW_add_lowerLambertW_eq_exp_branchGap`, `principalLambertW_add_lowerLambertW_eq_cosh_div_sinh_branchGap`, `principalLambertW_mul_lowerLambertW_eq_exp_branchGap`, `principalLambertW_mul_lowerLambertW_eq_sinh_sq_branchGap`, `principalLambertW_add_lowerLambertW_lt_neg_two`, `principalLambertW_mul_lowerLambertW_pos`, `principalLambertW_mul_lowerLambertW_lt_one`, and `principalLambertW_mul_lowerLambertW_mem_Ioo` (module `LambertWBranchSymmetry`)	Exact. The exponential-rational sum and product forms strengthen the printed hyperbolic identities, and the final theorem packages the two strict product bounds. The left endpoint is sharp: at $x = -1/\mathrm{e}$, both branches are -1, hence the sum is -2 and the product is 1; the open hypotheses are therefore essential. The endpoint $x = 0$ remains excluded because the lower real branch has no finite value there.
(16), with the standard removable-origin convention above, and (19): the complex Bernoulli generating identity and the compact paired series for $0 < \Delta < 2\pi$ on $-1/\mathrm{e} < x < 0$, together with the exact convergence locus	`summable_norm_bernoulli_mul_pow_div_factorial`, `summable_bernoulli_mul_pow_div_factorial_iff`, `hasSum_bernoulli_mul_pow_div_factorial`, `hasSum_bernoulli_mul_pow_div_factorial_complex_iff`, and `principalLambertW_lowerLambertW_eq_bernoulliSeries` (module `LambertWBranchGapBernoulli`)	Exact for the complex generating identity, the compact real branch pair, and the convergence locus.

Statement in this guide	Exact Lean declaration(s)	Remarks
Details for the preceding row. The first theorem proves absolute summability for every real z with $\lvert z\rvert < 2\pi$; the second proves, for complex z, that the Bernoulli series is summable if and only if $\lVert z\rVert < 2\pi$, hence it diverges on $\lVert z\rVert = 2\pi$ and throughout the exterior; the third proves the actual all-index real sum $z/(\exp z - 1)$ when $z \neq 0$; the fourth proves the complex HasSum value given by the inverse of `complexExpm1Div` at z if and only if $\lVert z\rVert < 2\pi$; and the fifth specializes both real sums on the strict common branch domain when the positive branch gap is below 2π. At $z = 0$, the fourth theorem's target is the removable value 1; away from zero it rewrites to the literal quotient. This makes both displayed identities in this row Exact under the stated convention, but claims no equality with Lean's totalized quotient at zero. The compiled radius proof uses even Bernoulli–zeta coefficient bounds and a positive even subsequence; it does not formalize the Guide's nearest-nonzero-zero explanation. The surrounding expanded displays and every Puiseux claim remain unpromoted.		
Theorem 11.4: for $-1/\mathrm{e} < x < 0$, $-1 + \sqrt{1 + \mathrm{e}x} < W_0(x) < -1 + \sqrt{2(1 + \mathrm{e}x)}$ and $W_0(x) < x$	`principalLambertW_negative_bounds`, with the halves `neg_one_add_sqrt_lt_principalLambertW`, `principalLambertW_lt_neg_one_add_sqrt`, `principalLambertW_lt_self_of_neg` (module `LambertWNegativeBounds`)	Exact. In the shifted variable $s = W_0 + 1 \in (0, 1)$ the two square-root bounds are the exponential inequalities $\mathrm{e}^s > 1 + s$ and $\mathrm{e}^s(1 - s) < 1 - s^2/2$; the second is `exp_mul_one_sub_lt_one_sub_half_sq`, proved from its derivative $s(\mathrm{e}^s - 1) > 0$, with no Lambert content.

Not formalized, in the order they appear above: the moments of Section 6, the local Taylor expansion, convergence of the Maclaurin series, the full Puiseux and logarithmic asymptotic expansions, the remaining bounds of Section 11 (the two-sided W_{-1} bound of Theorem 11.5), and everything from Section 12 onward.

G Concordance of the four merged treatments

This volume merges four independently written treatments that arrived together in the corpus inbox: the *spine* (the article whose body forms the main text above), the *monograph* treatment (`Lambert-W-Article/`), the *third* treatment (`Lambert_W_article/`), and the *fourth* treatment (`Lambert-W-Function-article-4/`). On the shared core the four agree theorem for theorem and constant for constant; the check below was performed before merging.

Topic	Status across the four treatments
Real branches, domains, curvature, inflection of W_{-1} at $-2/\mathrm{e}^2$	All four, identical statements and proofs.
Derivative polynomials P_n with recurrence $P_{n+1} = (1 + w)P_n' - (nw + 3n - 1)P_n$	All four; tables agree through $P_5 = 24w^4 + 192w^3 + 622w^2 + 974w + 625$.

Topic	Status across the four treatments
Maclaurin $(-n)^{n-1}/n!$, radius $1/\mathrm{e}$, tree function, Cayley	All four; the spine adds the endpoint convergence via Raabe.
Signed Puiseux series at $-1/\mathrm{e}$	All four; coefficients agree through $a_9 = 226287557/37623398400$; the fourth treatment adds the closed Lagrange form (140).
Asymptotic polynomials $\mathcal{P}_n(L)$ via unsigned Stirling cycle numbers	All four, identical through $n = 4$; identical unified two-branch statement.
Elementary bounds (principal positive, principal negative, Chatzigeorgiou W_{-1} bounds, near-zero logarithmic brackets)	All four, identical constants; the spine's midpoint corollary (85) is the sharpest packaged form.
Padé approximants and pole-freeness	Spine, monograph, third treatment; identical $R_{1,1}, R_{2,2}, R_{3,3}$.
Euler continued fraction = Maclaurin partial sums	All four, identical.
Contour/residue representation; Kalugin–Jeffrey–Corless cut integrals; complete monotonicity of W_0'	Spine in full; monograph and fourth treatment carry the contour and Fourier–logarithm forms.
Logarithmic Newton map, monotone global starts, residual certificates, Halley	All four; the spine's ordering theorem (Theorem 14.3) is the most complete.
Wright omega; $v - \log v = u$ two-branch geometry	All four.
Applications: Michaelis–Menten, diode, delay spectrum, SIR final size, primes	All four (with varying selections); complements add patch residence, Wien, linear drag, tortoise coordinate.
Lagrange–Bürmann proof, exercises with solutions, formula sheet	Spine and fourth treatment (independent write-ups; the spine's is retained).
Unique layers preserved in Section 21	Tower convergence theorem (monograph); inverse-Taylor corrections and seeds (monograph); fixed-point iteration criterion, toolkit, monodromy (third); $x^y = y^x$, involution, transcendence, scaling, unwinding, r-Lambert, extra applications, further-study list (fourth).

The four verification programs are preserved with their packages under `assets/` (three variants named `lambert_w_experiments.py` and one `numerical_experiments.py`); each independently validates the shared expansions, bounds, and iterations at high precision, and the spine's program regenerates every figure of this volume.

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